

Probability

OLIVER WEST

Contents

1	Fundamentals	2
1.1	Combinatorial analysis	3
1.2	Stirling's formula	4
2	Basic applications	9
2.1	Counting surjections	9
2.2	Counting derangements	10
3	Independence and conditional probability	11
4	Probability distributions	15
4.1	Discrete probability distributions	15
4.2	Random variables	17
4.3	Expectation	18
4.4	Moments, variance and covariance	22
4.5	Inequalities on random variables	26
4.6	Joint distribution and conditional expectation	29
5	Random walks	33
6	Probability generating functions	35
6.1	Sum of a random number of random variables	38
7	Branching processes	39
7.1	Extinction probability	40
8	Continuous random variables	43
8.1	The normal distribution	47
8.1.1	Linear transformations of a normal random variable	49
8.2	Multivariate density functions	50
8.2.1	Marginal density	51
8.2.2	Sum of independent random variables	52
8.2.3	Conditional density	52
8.3	Order statistics of a random sample	53
9	Moment generating functions	55
9.1	Multivariate moment generating functions	57
9.2	Multivariate Gaussian random variables	57
9.3	Bivariate Gaussians	61

10 Convergence results and limit theorems	62
10.1 Central limit theorem	65
11 Simulation of random variables	69
11.1 Box-Muller transform	69
11.2 Rejection sampling	70

Lecture 1 – 23/01/26

§1 Fundamentals

Definition 1.1. Let Ω be a set and $\mathcal{F}$ a collection of subsets of Ω . We call $\mathcal{F}$ a *σ -algebra* if

- $\Omega \in \mathcal{F}$;
- if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$;
- if $(A_n)_{n \in \mathbb{N}}$ is a countable collection of sets in $\mathcal{F}$, then

$$\bigcup A_n \in \mathcal{F}$$

Definition 1.2. Let $\mathcal{F}$ be a σ -algebra on Ω . A function

$$\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$$

is called a *probability measure* if

- $\mathbb{P}(\Omega) = 1$;
- for any countable disjoint collection $(A_n)_{n \in \mathbb{N}}$ in $\mathcal{F}$,

$$\mathbb{P}\left(\bigcup A_n\right) = \sum \mathbb{P}(A_n)$$

We call $\mathbb{P}(A)$ the *probability* of A .

Then $(\Omega, \mathcal{F}, \mathbb{P})$ is a *probability space*.

Definition 1.3. The elements of Ω are called *outcomes*, and the elements of the σ -algebra $\mathcal{F}$ are called *events*.

Importantly, we talk about probabilities of *events*, never probabilities of outcomes – the domain of $\mathbb{P}$ is $\mathcal{F}$, not Ω .

Remark. When Ω is countable, we take $\mathcal{F}$ to be all subsets of Ω . However, this framework will become useful when considering uncountable Ω .

Let us note some important properties of probability measures that follow from the definition.

- $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$;
- $\mathbb{P}(\emptyset) = 0$;

- $\mathbb{P}(A) \leq \mathbb{P}(B)$ if $A \subseteq B$;
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Example 1.4 • *Rolling a fair die.* Then $\Omega = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{F} = \mathcal{P}(\Omega)$, and

$$\mathbb{P}(A) = \frac{|A|}{6}$$

since all outcomes are equally likely.

- $\Omega = \{\omega_1, \dots, \omega_n\}$, $\mathcal{F} = \mathcal{P}(\Omega)$, $\mathbb{P}(A) = \frac{|A|}{|\Omega|}$ models the experiment of picking an element of Ω uniformly at random.
- *Picking balls from a bag.* Suppose we have n balls labelled $\{1, \dots, n\}$, and we pick $k \leq n$ balls uniformly at random without replacement. Then $\Omega = \{A : A \subseteq \{1, \dots, n\}, |A| = k\}$, so $|\Omega| = \binom{n}{k}$ and $\mathbb{P}(\{\omega\}) = \binom{n}{k}^{-1}$ for any $\omega \in \Omega$.
- *A deck of 52 cards.* Take a well-shuffled deck of 52 cards. Then $\Omega = \{\text{all permutations of 52 cards}\}$, so $|\Omega| = 52!$.

What is, for example, $\mathbb{P}(\text{top 2 cards are aces})$? It is $\frac{4 \cdot 3 \cdot 50!}{52!}$.

- *Largest digit.* Consider a string of n random digits from 0 up to 9. Then $\Omega = \{0, \dots, 9\}^n$, $|\Omega| = 10^n$.

Consider the events $A_k = \{\text{no digit exceeds } k\}$, $B_k = \{\text{largest digit is } k\}$. Then

$$\begin{aligned} \mathbb{P}(A_k) &= \frac{(k+1)^n}{10^n} \\ \implies \mathbb{P}(B_k) &= \mathbb{P}(A_k \setminus A_{k-1}) = \frac{(k+1)^n - k^n}{10^n} \end{aligned}$$

- *Birthday problem.* There are n people. What is the probability that at least 2 of them share the same birthday?

Assume each birthday is equally likely to be one of $\{1, \dots, 365\}$. Then $\Omega = \{1, \dots, 365\}^n$, $|\Omega| = 365^n$, and $\mathbb{P}(\{\omega\}) = \frac{1}{365^n}$.

Let $A = \{\text{at least 2 people have the same birthday}\}$. Instead we count $A^c = \{\text{all birthdays different}\}$.

$$\begin{aligned} \mathbb{P}(A^c) &= \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n} \\ \implies \mathbb{P}(A) &= 1 - \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n} \end{aligned}$$

Perhaps surprisingly, if $n = 22$, $\mathbb{P}(A) = 0.476$, while if $n = 23$, $\mathbb{P}(A) = 0.507$.

§1.1 Combinatorial analysis

Let us introduce some notions.

1. Let Ω be a finite set with $|\Omega| = n$. We want to partition Ω into k disjoint subsets $\Omega_1, \dots, \Omega_k$ with $|\Omega_i| = n_i$ and $\sum n_i = n$. How many ways are there to do this? It is

$$\begin{aligned} M &= \binom{n}{n_1} \cdot \binom{n-n_1}{n_2} \cdots \binom{n-(n_1+\cdots+n_{k-1})}{n_k} \\ &= \frac{n!}{n_1!n_2!\cdots n_k!} \\ &= \binom{n}{n_1, n_2, \dots, n_k} \end{aligned}$$

This is the *multinomial coefficient*.

2. Suppose $f : \{1, \dots, k\} \rightarrow \{1, \dots, n\}$ (with $k \leq n$). f is *strictly increasing* if $x < y \implies f(x) < f(y)$, and f is *increasing* if $x < y \implies f(x) \leq f(y)$.

How many strictly increasing $f : \{1, \dots, k\} \rightarrow \{1, \dots, n\}$ are there? Such a function is determined uniquely by its range, so there are $\binom{n}{k}$ of them.

What about increasing functions? Let's define a bijection

$$\{f : \{1, \dots, k\} \rightarrow \{1, \dots, n\} \text{ inc.}\} \rightarrow \{f : \{1, \dots, k\} \rightarrow \{1, n+k-1\} \text{ strictly inc.}\}$$

which takes f to g with $g(i) = f(i) + i - 1$. Hence we have $\binom{n+k-1}{k}$ such functions.

Definition 1.5 (Asymptotic notation). If (a_n) and (b_n) are sequences, we write $a_n \sim b_n$ if

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 1$$

§1.2 Stirling's formula

Theorem 1.6 (Stirling)

$$n! \sim n^n \sqrt{2\pi n} e^{-n}$$

as $n \rightarrow \infty$.

Weaker statement. We shall first show a weaker statement, that $\log n! \sim n \log n$.

Define $l_n = \log n! = \log 2 + \cdots + \log n$. We have

$$\log \lfloor x \rfloor \leq \log x \leq \log \lceil x \rceil$$

Integrating from 1 to n gives

$$\begin{aligned} \sum_{k=1}^{n-1} \log k &\leq \int_1^n \log x \, dx \leq \sum_{k=1}^n \log k \\ \implies l_{n-1} &\leq n \log n - n + 1 \leq l_n \\ \implies n \log n - n + 1 &\leq l_n \leq (n+1) \log(n+1) - (n+1) + 1 \end{aligned}$$

Now the result follows by dividing through by $n \log n$ and taking the limit. ■

Non-examinable proof of Stirling. For every twice-differentiable $f : \mathbb{R} \rightarrow \mathbb{R}$, for all $a < b$, we have

$$\int_a^b f(x) dx = \frac{f(a) + f(b)}{2}(b - a) - \frac{1}{2} \int_a^b (x - a)(b - x) f''(x) dx$$

obtained by integrating twice by parts. Put $f = \log$, $a = k$, $b = k + 1$ for $k \in \mathbb{N}$, to get

$$\int_k^{k+1} \log x dx = \frac{\log k + \log(k + 1)}{2} + \frac{1}{2} \int_k^{k+1} \frac{(x - k)(k + 1 - x)}{x^2} dx$$

Summing from $k = 1$ to $n - 1$, we have

$$\int_1^n \log x dx = \frac{\log(n - 1)! + \log n!}{2} + \sum_{k=1}^{n-1} a_k$$

where $a_k = \frac{1}{2} \int_0^1 \frac{x(1-x)}{(x+k)^2} dx$. Hence

$$n \log n - n + 1 = \log n! - \frac{\log n}{2} + \sum_{k=1}^{n-1} a_k$$

and so

$$\begin{aligned} \log n! &= n \log n - n + 1 + \frac{\log n}{2} - \sum_{k=1}^{n-1} a_k \\ \implies n! &= n^n \cdot e^{-n} \cdot \sqrt{n} \cdot \exp \left(1 - \sum_{k=1}^{n-1} a_k \right) \end{aligned}$$

Let us now consider

$$\begin{aligned} a_k &= \frac{1}{2} \int_0^1 \frac{x(1-x)}{(x+k)^2} dx \\ &\leq \frac{1}{2} \int_0^1 \frac{1}{2k^2} \int_0^1 x(1-x) dx \\ &= \frac{1}{12k^2} \end{aligned}$$

and so $\sum_{k=1}^{\infty} a_k < \infty$. Now put

$$\begin{aligned} A &= \exp \left(1 - \sum_{k=1}^{\infty} a_k \right) \\ \implies n! &= n^n \cdot e^{-n} \cdot \sqrt{n} \cdot A \cdot \underbrace{\exp \left(\sum_{k=n}^{\infty} a_k \right)}_{\rightarrow 1 \text{ as } n \rightarrow \infty} \\ \implies \frac{n!}{n^n e^{-n} \sqrt{n}} &\rightarrow A \text{ as } n \rightarrow \infty \end{aligned}$$

It remains to show that $A = \sqrt{2\pi}$. Consider

$$2^{-2n} \binom{2n}{n} = 2^{-2n} \frac{(2n)!}{(n!)^2}$$

$$\begin{aligned} &\sim \frac{2^{-2n} (2n)^{2n} e^{-2n} \sqrt{2n} A}{n^{2n} e^{-2n} n A^2} \\ &= \frac{\sqrt{2}}{A\sqrt{n}} \end{aligned}$$

Now, we shall use a different method to show

$$2^{-2n} \binom{2n}{n} \sim \frac{1}{\sqrt{\pi n}} \text{ as } n \rightarrow \infty$$

Consider

$$I_n = \int_0^{\pi/2} \cos^n \theta \, d\theta$$

We have $I_0 = \frac{\pi}{2}$, $I_1 = 1$, and, by integrating by parts, we have

$$I_n = \frac{n-1}{n} I_{n-2} \quad (*)$$

so that

$$\begin{aligned} I_{2n} &= \frac{2n-1}{2n} I_{2n-2} \\ &= \cdots = \frac{(2n-1)(2n-3)\cdots 3 \cdot 1}{(2n)(2n-2)\cdots 2} \cdot \frac{\pi}{2} \\ &= \frac{2n(2n-1)(2n-2)\cdots 3 \cdot 2 \cdot 1}{2n \cdot 2n \cdot (2n-2) \cdot (2n-2) \cdots 2 \cdot 2} \cdot \frac{\pi}{2} \\ &= \frac{(2n)!}{2^{2n} (n!)^2} \cdot \frac{\pi}{2} \\ &= \frac{\pi}{2} \cdot 2^{-2n} \binom{2n}{n} \end{aligned}$$

and we can similarly obtain

$$\begin{aligned} I_{2n+1} &= \frac{2n \cdots 4 \cdot 2}{(2n+1) \cdots 3 \cdot 1} \cdot 1 \\ &= \frac{1}{2n+1} \left(2^{-2n} \binom{2n}{n} \right)^{-1} \end{aligned}$$

Now, we want to show that $\frac{I_{2n}}{I_{2n+1}} \rightarrow 1$ as $n \rightarrow \infty$. By (*), we have

$$\frac{I_n}{I_{n-2}} \rightarrow 1 \text{ as } n \rightarrow \infty$$

But $I_n = \int_0^{\pi/2} \cos^n \theta \, d\theta$, so I_n is decreasing in n . This gives

$$\frac{I_{2n}}{I_{2n-2}} \leq \frac{I_{2n}}{I_{2n+1}} \leq \frac{I_{2n-1}}{I_{2n+1}}$$

which shows the result by sandwich. Hence, we get, as $n \rightarrow \infty$,

$$\begin{aligned} &\frac{\frac{\pi}{2} \cdot 2^{-2n} \cdot \binom{2n}{n}}{\frac{1}{2n+1} \cdot \left(2^{-2n} \binom{2n}{n} \right)^{-1}} \rightarrow 1 \\ \implies &\left(2^{-2n} \binom{2n}{n} \right)^2 \sim \frac{1}{\pi n} \end{aligned}$$

■

Proposition 1.7 (Countable subadditivity)

Let (A_n) be a collection of events in $\mathcal{F}$. Then

$$\mathbb{P}\left(\bigcup A_n\right) \leq \sum \mathbb{P}(A_n)$$

Proof. Let $B_1 = A_1$, and for $n \geq 2$, let $B_n = A_n \setminus (A_1 \cup \cdots \cup A_{n-1})$. By definition, it's immediate that (B_n) is a disjoint collection of sets in $\mathcal{F}$, and $\bigcup B_n = \bigcup A_n$, so

$$\begin{aligned} \mathbb{P}\left(\bigcup A_n\right) &= \mathbb{P}\left(\bigcup B_n\right) = \sum \mathbb{P}(B_n) \\ &\leq \sum \mathbb{P}(A_n) \end{aligned}$$

since $B_n \subseteq A_n$ for all n . ■

Proposition 1.8 (Continuity of probability measures)

Take a sequence of events $(A_n)_{n \in \mathbb{N}}$ such that $A_n \subseteq A_{n+1}$ for all n – we say (A_n) is an *increasing* sequence.

Then $\mathbb{P}(A_n)$ is increasing and converges. In particular,

$$\mathbb{P}(A_n) \rightarrow \mathbb{P}\left(\bigcup A_n\right)$$

as $n \rightarrow \infty$.

Proof. Define $B_1 = A_1$, and for $n \geq 2$, let $B_n = A_n \setminus (A_1 \cup \cdots \cup A_{n-1})$. Then

$$\bigcup_{k=1}^n B_k = A_n$$

and the B_n 's are disjoint. Then

$$\begin{aligned} \mathbb{P}(A_n) &= \mathbb{P}\left(\bigcup_{k=1}^n B_k\right) \\ &= \sum_{k=1}^n \mathbb{P}(B_k) \rightarrow \sum_{k=1}^{\infty} \mathbb{P}(B_k) \end{aligned}$$

Also,

$$\begin{aligned} \sum_{k=1}^{\infty} \mathbb{P}(B_k) &= \mathbb{P}\left(\bigcup B_k\right) \\ &= \mathbb{P}(A_k) \end{aligned}$$

■

Corollary 1.9

Analogously, take a sequence of events (A_n) such that $A_{n+1} \subseteq A_n$ for all n – we say (A_n) is a decreasing sequence. Then

$$\mathbb{P}(A_n) \rightarrow \mathbb{P}\left(\bigcap A_n\right)$$

Proof. Take complements and use the previous claim. ■

Proposition 1.10 (Inclusion-exclusion formula for probability measures)

Let $A_1, \dots, A_n \in \mathcal{F}$. Then

$$\mathbb{P}\left(\bigcup A_i\right) = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k})$$

Proof. We proceed by induction.

Base case. ADD

Inductive step. We have

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^n A_i\right) &= \mathbb{P}\left(\left(\bigcup_{i=1}^{n-1} A_i\right) \cup A_n\right) \\ &= \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\left(\bigcup_{i=1}^{n-1} A_i\right) \cap A_n\right) \\ &= \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\bigcup_{i=1}^{n-1} \underbrace{(A_i \cap A_n)}_{B_i}\right) \end{aligned}$$

Applying the inductive hypothesis, we get

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) &= \sum_{k=1}^{n-1} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) \\ \mathbb{P}\left(\bigcup_{i=1}^{n-1} B_i\right) &= \sum_{k=1}^{n-1} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k} \cap A_n) \end{aligned}$$

and subbing this in, we will find that everything works out. ■

Proposition 1.11 (Bonferroni inequalities)

Let $A_1, \dots, A_n$ be events. Then

$$\mathbb{P}\left(\bigcup A_i\right) \begin{cases} \leq \sum_{k=1}^r (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) & \text{if } r \text{ odd} \\ \geq \sum_{k=1}^r (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) & \text{if } r \text{ even} \end{cases}$$

where those sums are just truncated expansions of inclusion-exclusion.

Proof. We induct again.

Base case. $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \leq \mathbb{P}(A) + \mathbb{P}(B)$

Inductive step. Suppose r is odd. Then

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\bigcup_{i=1}^{n-1} \underbrace{(A_i \cap A_n)}_{B_i}\right)$$

By the inductive hypotheses,

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) &\leq \sum_{k=1}^r (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) \\ \mathbb{P}\left(\bigcup_{i=1}^{n-1} (A_i \cap A_n)\right) &\geq \sum_{k=1}^{r-1} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k} \cap A_n) \end{aligned}$$

where in the former, we choose to apply the identity for odd r , and in the latter, we apply it for even $r-1$. Substituting much as for the proof of the inclusion-exclusion formula, we obtain the result. The even case is analogous. $\blacksquare$

Let us now consider the special case where Ω is a finite set and $\mathbb{P}(A) = |A|/|\Omega|$. Then we have

Corollary 1.12 (Inclusion-exclusion principle for sets)

Suppose Ω is finite. Then

$$|A_1 \cup \dots \cup A_n| = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} |A_{i_1} \cap \dots \cap A_{i_k}|$$

Proof. Consider the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ where $\mathbb{P}(A) = |A|/|\Omega|$. Then, apply inclusion-exclusion and cancel $1/|\Omega|$ from both sides. $\blacksquare$

§2 Basic applications

§2.1 Counting surjections

Define a probability space as follows:

$$\Omega = \{f : \{1, \dots, n\} \rightarrow \{1, \dots, m\}\}$$

$$A = \{f \in \Omega : f \text{ is a surjection}\}$$

What is $|A|$? Let us define a set of events

$$A_i = \{f \in \Omega : i \notin \{f(1), \dots, f(n)\}\}, i \in \{1, \dots, m\}$$

so that

$$\begin{aligned} A &= A_1^c \cap \dots \cap A_m^c = (A_1 \cup \dots \cup A_m)^c \\ \implies |A| &= |\Omega| - |A_1 \cup \dots \cup A_m| \\ &= |\Omega| - \sum_{k=1}^m (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq m} |A_{i_1} \cap \dots \cap A_{i_k}| \\ &= \underbrace{|\Omega|}_{m^n} - \sum_{k=1}^m (-1)^{k+1} \binom{m}{k} (m-k)^n \\ &= \sum_{k=0}^m (-1)^k \binom{m}{k} (m-k)^n \end{aligned}$$

Lecture 4 – 30/01/26

§2.2 Counting derangements

Definition 2.1. A *derangement* is a permutation with no fixed points.

Let

$$\begin{aligned} \Omega &= \{\text{permutations of } \{1, \dots, n\}\} \\ A &= \{\text{derangements}\} \end{aligned}$$

Let

$$A_i = \{f \in \Omega : f(i) = i\}$$

so that

$$\begin{aligned} A &= A_1^c \cap \dots \cap A_n^c = (A_1 \cup \dots \cup A_n)^c \\ \implies \mathbb{P}(A) &= 1 - \mathbb{P}\left(\bigcup A_i\right) = 1 - \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \underbrace{\mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k})}_{\frac{(n-k)!}{n!}} \\ &= 1 - \sum_{k=1}^n (-1)^{k+1} \binom{n}{k} \frac{(n-k)!}{n!} \\ &= 1 - \sum_{k=1}^n (-1)^{k+1} \frac{1}{k!} \\ &= \sum_{k=0}^n \frac{(-1)^k}{k!} \rightarrow e^{-1} \end{aligned}$$

§3 Independence and conditional probability

Definition 3.1 (Independence). Let $A, B \in \mathcal{F}$. We say A and B are *independent* and write $A \perp B$ if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$$

Likewise, a countable collection of events $(A_n)_{n \in \mathbb{N}}$ is said to be independent if, for distinct $i_1, \dots, i_k$,

$$\mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) = \prod_{j=1}^k \mathbb{P}(A_{i_j})$$

Remark. Pairwise independence of events does not imply independence.

Example

Toss a fair coin twice. We have

$$\begin{aligned} \Omega &= \{(0, 0), (0, 1), (1, 0), (1, 1)\} \\ \mathbb{P}(\{\omega\}) &= \frac{1}{4} \end{aligned}$$

Let

$$\begin{aligned} A &= \{(0, 0), (0, 1)\} \\ B &= \{(0, 0), (1, 0)\} \\ C &= \{(1, 0), (0, 1)\} \end{aligned}$$

We have $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C)$, and furthermore we can check that $A \perp B$, $A \perp C$ and $B \perp C$. But $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(\emptyset) = 0$, so they are not independent.

Proposition 3.2

If A is independent of B , then A is independent of B^c .

Proof. We have

$$\begin{aligned} \mathbb{P}(A \cap B^c) &= \mathbb{P}(A) - \mathbb{P}(A \cap B) \\ &= \mathbb{P}(A)(1 - \mathbb{P}(B)) \\ &= \mathbb{P}(A)\mathbb{P}(B^c) \end{aligned}$$

■

Definition 3.3 (Conditional probability). Take a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$. For $A \in \mathcal{F}$ we write the *conditional probability of A given B*

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

Remark. Notably, if $A \perp B$, then $\mathbb{P}(A|B) = \mathbb{P}(A)$, as we would expect.

Proposition 3.4 (Countable additivity for conditional probability)

Suppose $(A_n)_{n \in \mathbb{N}}$ is a disjoint collection in $\mathcal{F}$. Then

$$\mathbb{P}\left(\bigcup A_n | B\right) = \sum \mathbb{P}(A_n | B)$$

Proof. We have

$$\begin{aligned} \mathbb{P}\left(\bigcup A_n | B\right) &= \frac{\mathbb{P}(\bigcup A_n \cap B)}{\mathbb{P}(B)} \\ &= \frac{\mathbb{P}(\bigcup (A_n \cap B))}{\mathbb{P}(B)} \\ &= \sum \frac{\mathbb{P}(A_n \cap B)}{\mathbb{P}(B)} \\ &= \sum \mathbb{P}(A_n | B) \end{aligned}$$

■

Proposition 3.5 (Law of total probability)

Suppose $(B_n)_{n \in \mathbb{N}}$ is a disjoint collection in $\mathcal{F}$, $\bigcup B_n = \Omega$, and $\mathbb{P}(B_n) > 0$ for every n . Let $A \in \mathcal{F}$, then

$$\mathbb{P}(A) = \sum_n \mathbb{P}(A | B_n) \mathbb{P}(B_n)$$

Proof. We have

$$\begin{aligned} \mathbb{P}(A) &= \mathbb{P}\left(A \cap \left(\bigcup B_n\right)\right) \\ &= \mathbb{P}\left(\bigcup (B_n \cap A)\right) \\ &= \sum \mathbb{P}(B_n \cap A) \\ &= \sum \mathbb{P}(A | B_n) \mathbb{P}(B_n) \end{aligned}$$

■

Proposition 3.6 (Bayes' theorem)

Suppose (B_n) are disjoint, $\bigcup B_n = \Omega$ and $\mathbb{P}(B_n) > 0$ for every n . Let $A \in \mathcal{F}$, then

$$\mathbb{P}(B_n | A) = \frac{\mathbb{P}(A | B_n) \cdot \mathbb{P}(B_n)}{\sum_k \mathbb{P}(A | B_k) \mathbb{P}(B_k)}$$

Proof sketch. Use the definitions of $\mathbb{P}(B_n|A)$ and $\mathbb{P}(A|B_n)$, and expand the bottom using the law of total probability. ■

This formula is the basis of so-called Bayesian statistics. We know the probability of $\mathbb{P}(B_n)$ and we have a model which gives us $\mathbb{P}(A|B_n)$. With the formula, we find the *posterior* probability of B_n given that A occurred.

Example (False positives for a rare condition)

Suppose that condition A affects 0.1% of the population. We have a medical test which is positive for 98% of the affected population and 1% of the unaffected population. What is the probability that a random individual who tests positive has the disease?

Put $A = \{\text{individual suffers from } A\}$, $P = \{\text{individual tests positive}\}$. We have $\mathbb{P}(A) = 0.001$, $\mathbb{P}(P|A) = 0.98$, $\mathbb{P}(P|A^c) = 0.01$, so

$$\begin{aligned}\mathbb{P}(A|P) &= \frac{\mathbb{P}(P|A)\mathbb{P}(A)}{\mathbb{P}(P|A)\mathbb{P}(A) + \mathbb{P}(P|A^c)\mathbb{P}(A^c)} \\ &= 0.089\end{aligned}$$

In particular, we can see

$$\mathbb{P}(A|P) = \frac{1}{1 + \frac{\mathbb{P}(P|A^c)\mathbb{P}(A^c)}{\mathbb{P}(P|A)\mathbb{P}(A)}}$$

Since $\mathbb{P}(A^c)$ and $\mathbb{P}(P|A)$ are both very close to one, the relevant ratio is

$$\frac{\mathbb{P}(P|A^c)}{\mathbb{P}(A)}$$

But since $\mathbb{P}(P|A^c) \gg \mathbb{P}(A)$, this causes a very small value for $\mathbb{P}(A|P)$.

Example • What is the probability that I have two boys given that I have two children one of whom is a boy?

Since no further information is given, we take all outcomes to be equally likely. Let

$$BG = \{\text{elder child a boy, younger a girl}\}$$

with GB, BB, GG defined analogously. Then we want

$$\begin{aligned}\mathbb{P}(BB|BB \cup BG \cup GB) &= \frac{\mathbb{P}(BB)}{\mathbb{P}(BB \cup BG \cup GB)} \\ &= \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}\end{aligned}$$

- What about $\mathbb{P}(\text{both boys}|\text{eldest a boy})$? This is

$$\mathbb{P}(BB|BB \cup BG) = \frac{1}{2}$$

- What about $\mathbb{P}(\text{both boys}|\text{one a boy born on a Thursday})$? Denote a boy born on Thursday by T, a boy not born on Thursday by N, and a girl by G. Then we want

$$\begin{aligned}&\mathbb{P}(TT \cup TN \cup NT|TT \cup TG \cup GT \cup TN \cup NT) \\ &= \frac{\mathbb{P}(TT \cup TN \cup NT)}{\mathbb{P}(TT \cup TG \cup GT \cup TN \cup NT)} \\ &= \frac{\left(\frac{1}{2} \cdot \frac{1}{7}\right)^2 + \left(\frac{1}{2} \cdot \frac{1}{7}\right)\left(\frac{1}{2} \cdot \frac{6}{7}\right) + \left(\frac{1}{2} \cdot \frac{6}{7}\right)\left(\frac{1}{2} \cdot \frac{1}{7}\right)}{\left(\frac{1}{2} \cdot \frac{1}{7}\right)^2 + \left(\frac{1}{2} \cdot \frac{1}{7}\right) \cdot \frac{1}{2} + \frac{1}{2} \cdot \left(\frac{1}{2} \cdot \frac{1}{7}\right) + \left(\frac{1}{2} \cdot \frac{1}{7}\right)\left(\frac{1}{2} \cdot \frac{6}{7}\right) + \left(\frac{1}{2} \cdot \frac{6}{7}\right)\left(\frac{1}{2} \cdot \frac{1}{7}\right)} \\ &= \frac{13}{27}\end{aligned}$$

Morally speaking, we're providing information that helps specify the boy in question more, which brings the probability closer to a half.

Example (Simpson's paradox)

Consider the following data:

<i>All applicants</i>	Admitted	Rejected	% Admitted
State	25	25	50%
Indep.	28	22	56%
<i>London schools</i>	Admitted	Rejected	% Admitted
State	15	22	41%
Indep.	5	8	38%

<i>Cambridge schools</i>	Admitted	Rejected	% Admitted
State	10	3	77%
Indep.	23	14	62%

– paradoxically, the state admission rate is higher when the school locations are taken individually, but the independent admission right is higher for the combined data. This phenomenon is called *confounding*, and arises when we aggregate data from disparate populations.

Let

$$\begin{aligned} A &= \{\text{individual is admitted}\} \\ B &= \{\text{individual from London}\} \\ C &= \{\text{individual from state school}\} \end{aligned}$$

Then

$$\begin{aligned} \mathbb{P}(A|B \cap C) &> \mathbb{P}(A|B \cap C^c) \\ \mathbb{P}(A|B^c \cap C) &> \mathbb{P}(A|B^c \cap C^c) \\ \text{but } \mathbb{P}(A|C) &< \mathbb{P}(A|C^c) \end{aligned}$$

Let's examine how this can happen. We have

$$\begin{aligned} \mathbb{P}(A|C) &= \mathbb{P}(A \cap B|C) + \mathbb{P}(A \cap B^c|C) \\ &= \frac{\mathbb{P}(A \cap B \cap C)}{\mathbb{P}(C)} + \frac{\mathbb{P}(A \cap B^c \cap C)}{\mathbb{P}(C)} \\ &= \frac{\mathbb{P}(A|B \cap C)\mathbb{P}(B \cap C)}{\mathbb{P}(C)} + \frac{\mathbb{P}(A|B^c \cap C)\mathbb{P}(B^c \cap C)}{\mathbb{P}(C)} \\ &= \mathbb{P}(A|B \cap C)\mathbb{P}(B|C) + \mathbb{P}(A|B^c \cap C)\mathbb{P}(B^c|C) \\ &> \mathbb{P}(A|B \cap C^c)\mathbb{P}(B|C) + \mathbb{P}(A|B^c \cap C^c)\mathbb{P}(B^c|C) \end{aligned}$$

by the first two inequalities from above. Now, if $\mathbb{P}(B|C) = \mathbb{P}(B|C^c)$, then we would get $\mathbb{P}(A|C) > \mathbb{P}(A|C^c)$, which is the intuitive expectation – but here this does not hold, and so we can make the ‘paradox’ hold.

Even more simply, this is just the fact that, if $\frac{a_1}{a_2} > \frac{b_1}{b_2}$ and $\frac{c_1}{c_2} > \frac{d_1}{d_2}$, it does not necessarily hold that $\frac{a_1+c_1}{a_2+c_2} > \frac{b_1+d_1}{b_2+d_2}$.

§4 Probability distributions

§4.1 Discrete probability distributions

Take a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with Ω either finite or countable – that is, $\Omega = \{\omega_1, \omega_2, \dots\}$. As normal, we take $\mathcal{F} = \mathcal{P}(\Omega)$, and so we need only care about $\mathbb{P}(\{\omega_i\})$ for every i .

Definition 4.1 (Discrete probability distributions). For such a setup, we call the sequence of probabilities $(\mathbb{P}(\{\omega_i\}))_i$ a *discrete probability distribution*. We may also write $p_i = \mathbb{P}(\{\omega_i\})$.

Let us consider some examples of discrete probability distributions.

- ① *Bernoulli distribution.* The Bernoulli distribution models the outcome of a toss of a biased coin. We have $\Omega = \{0, 1\}$, $p_0 = \mathbb{P}(\{0\}) = 1 - p$, $p_1 = \mathbb{P}(\{1\}) = p$ for some p with $p \in [0, 1]$.
- ② *Binomial distribution.* We denote this $\text{Bin}(n, p)$, with $n \in \mathbb{N}$, $p \in [0, 1]$. It models the number of heads seen for n independent tosses of a p -biased coin. We have

$$\Omega = \{0, 1, \dots, n\}$$

$$p_k = \binom{n}{k} p^k (1 - p)^{n-k}$$

- ③ *Multinomial distribution.* We denote this $M(n, p_1, \dots, p_k)$, where $p_i \geq 0$, $\sum p_i = 1$. Given k boxes and n balls placed in box i with p_i , the multinomial distribution gives the probability of n_l balls in box l for every n . We have

$$\Omega = \left\{ (n_1, \dots, n_k) \in \mathbb{N}^k : \sum_{i=1}^k n_i = n \right\}$$

$$\mathbb{P}(n_1, \dots, n_k) = \binom{n}{n_1} p_1^{n_1} \cdot \binom{n - n_1}{n_2} p_2^{n_2} \cdots \binom{n - n_1 - \dots - n_{k-1}}{n_k} p_k^{n_k}$$

$$= \binom{n}{n_1, n_2, \dots, n_k} p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k}$$

- ④ *Geometric distribution.* The geometric distribution gives the probability that the first heads achieved on a p -coin is after k tosses:

$$\Omega = \{1, 2, \dots\}$$

$$p_k = (1 - p)^{k-1} p$$

Alternatively, we might want to model the probability that we see k tails before the first heads:

$$\Omega = \{0, 1, \dots\}$$

$$p'_k = (1 - p)^k p$$

- ⑤ *Poisson distribution.* Suppose customers arrive into a shop in $[0, 1]$. Discretise $[0, 1]$ into $\left\{ \left[\frac{i-1}{N}, \frac{i}{N} \right] \right\}$, $i = 1, \dots, N$, $N \in \mathbb{N}$. Independently, in each interval, a customer arrives with probability p , and doesn't arrive with probability $1 - p$.

Then

$$\mathbb{P}(k \text{ customers have arrived}) = \binom{N}{k} p^k (1 - p)^{N-k}$$

– in other words, this follows a binomial distribution. Now, take $p = \frac{\lambda}{N}$, so

$$\begin{aligned} \mathbb{P}(k \text{ customers have arrived}) &= \frac{N!}{k!(N-k)!} \left(\frac{\lambda}{N} \right)^k \left(1 - \frac{\lambda}{N} \right)^{N-k} \\ &= \frac{\lambda^k}{k!} \cdot \underbrace{\frac{N(N-1)\dots(N-k+1)}{N^k}}_{\rightarrow 1} \underbrace{\left(1 - \frac{\lambda}{N} \right)^{N-k}}_{\rightarrow e^{-\lambda}} \end{aligned}$$

$$\rightarrow e^{-\lambda} \frac{\lambda^k}{k!}$$

as $N \rightarrow \infty$. Hence we have

$$\begin{aligned}\Omega &= \{0, 1, 2, \dots\} \\ p_k &= e^{-\lambda} \frac{\lambda^k}{k!}\end{aligned}$$

and we denote this $\text{Poi}(\lambda)$.

§4.2 Random variables

Definition 4.2. Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a *random variable* is a function

$$X : \Omega \rightarrow \mathbb{R}$$

satisfying

$$\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}$$

for any x . We have the shorthand notation $\{X \leq x\}$ to refer to $\{\omega \in \Omega : X(\omega) \leq x\}$, and $\{X \in A\} = \{\omega \in \Omega : X(\omega) \in A\}$.

Definition 4.3. Let $A \in \mathcal{F}$. Define the *indicator* of A

$$1_A(\omega) = 1(\omega \in A) = \begin{cases} 1 & \omega \in A \\ 0 & \omega \notin A \end{cases}$$

Since $A \in \mathcal{F}$, 1_A is a random variable.

Definition 4.4. Suppose X is a random variable. The *probability distribution function* of X is given by

$$\begin{aligned}F_X(x) &: \mathbb{R} \rightarrow [0, 1] \\ x &\mapsto \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq x\}) = \mathbb{P}(X \leq x)\end{aligned}$$

Definition 4.5. $X = (X_1, \dots, X_n)$ is called a *random variable in $\mathbb{R}^n$* if

$$(X_1, \dots, X_n) : \Omega \rightarrow \mathbb{R}^n$$

is a function such that, for any $x_1, \dots, x_n$,

$$\{\omega : X_1(\omega) \leq x_1, \dots, X_n(\omega) \leq x_n\} \in \mathcal{F}$$

which we denote with the shorthand $\{X_1 \leq x_1, \dots, X_n \leq x_n\}$.

Note that $(X_1, \dots, X_n)$ is a random variable if and only if all of $X_1, \dots, X_n$ are.

Definition 4.6. A random variable X is called *discrete* if it takes values in some countable set S .

Definition 4.7. For a discrete random variable, we are able to write

$$p_x = \mathbb{P}(X = x) = \mathbb{P}(\omega : X(\omega) = x)$$

Then, $(p_x)_{x \in S}$ is called the *probability mass function* of X .

Now we are able to say things like

$$\text{for any } A \subseteq S, \mathbb{P}(X \in A) = \sum_{x \in A} p_x$$

If p_x follows a given distribution, then we say X is a random variable with that distribution or a random variable following that distribution.

Definition 4.8. Let $X_1, \dots, X_n$ be discrete random variables with values in $S_1, \dots, S_n$. They are called *independent* if

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \mathbb{P}(X_1 = x_1) \dots \mathbb{P}(X_n = x_n)$$

for all $x_1 \in S_1, \dots, x_n \in S_n$.

Example 4.9

Toss a p -coin N times independently. Then

$$\Omega = \{0, 1\}^N$$

and every $\omega \in \Omega$ is of the form $(\omega_1, \dots, \omega_N)$, where ω_i is the i^{th} outcome. Hence

$$p_\omega = \mathbb{P}(\{\omega\}) = \prod_{k=1}^N p^{\omega_k} (1-p)^{1-\omega_k}$$

Let us also define random variables X_k for each k , satisfying $X_k(\omega) = \omega_k$. Clearly X_k is a Bernoulli r.v. of parameter p . We claim $X_1, \dots, X_N$ are independent: we have

$$\begin{aligned} P(X_1 = x_1, \dots, X_N = x_N) &= \prod_{k=1}^N p^{x_k} (1-p)^{1-x_k} \\ &= \prod_{k=1}^N \mathbb{P}(X_k = x_k) \end{aligned}$$

Lecture 7 – 06/02/26

§4.3 Expectation

Definition 4.10. Suppose Ω is finite or countable, and let $X : \Omega \rightarrow \mathbb{R}$ be a discrete random variable. We say X is *non-negative* if $X \geq 0$ – then we define the *expectation* of X by

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} \mathbb{P}(\{\omega\}) \cdot X(\omega)$$

We can equivalently let

$$\Omega_X = \{X(\omega) : \omega \in \Omega\}$$

so that

$$\begin{aligned}
 \mathbb{E}[X] &= \sum_{\omega \in \Omega} X(\omega) \cdot \mathbb{P}(\{\omega\}) \\
 &= \sum_{x \in \Omega_X} \sum_{\omega \in \{X=x\}} X(\omega) \mathbb{P}(\{\omega\}) \\
 &= \sum_{x \in \Omega_X} \sum_{\omega \in \{X=x\}} x \mathbb{P}(\{\omega\}) \\
 &= \sum_{x \in \Omega_X} x \sum_{\omega \in \{X=x\}} \mathbb{P}(\{\omega\}) \\
 &= \sum_{x \in \Omega_X} x \cdot \mathbb{P}(X = x)
 \end{aligned}$$

or in other words the average of the values taken by X , weighted by their probabilities.

Example 4.11

Suppose $X \sim \text{Bin}(n, p)$. Then

$$\begin{aligned}
 \mathbb{E}[X] &= \sum_{k=0}^N k \mathbb{P}(X = k) \\
 &= \sum_{k=0}^N k \binom{N}{k} p^k (1-p)^{N-k} \\
 &= \sum_{k=0}^N k \frac{N!}{(N-k)!k!} p^k (1-p)^{N-k} \\
 &= \sum_{k=0}^{N-1} N \cdot \binom{N-1}{k} p^{k+1} (1-p)^{N-k-1} \cdot p \\
 &= Np \cdot (p + 1 - p)^{N-1} = Np
 \end{aligned}$$

Example 4.12

Suppose $X \sim \text{Poi}(\lambda)$. Then

$$\begin{aligned}
 \mathbb{E}[X] &= \sum_{k=0}^{\infty} k \mathbb{P}(X = k) \\
 &= \sum_{k=0}^{\infty} k e^{-\lambda} \frac{\lambda^k}{k!} \\
 &= e^{-\lambda} \cdot \lambda \sum_{k=0}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda
 \end{aligned}$$

Let X be a random variable – then we define

$$\begin{aligned}
 X^+ &= \max(X, 0) \\
 X^- &= \max(-X, 0)
 \end{aligned}$$

so $X = X^+ - X^-$ and $|X| = X^+ + X^-$. Supposing X is discrete, we can define the expectations of $\mathbb{E}[X^+]$ and $\mathbb{E}[X^-]$, since they are both non-negative.

If at least one of $\mathbb{E}[X^+]$ and $\mathbb{E}[X^-]$ is finite, we define

$$\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-]$$

If both are infinite, the expectation of X is undefined.

Also, if $\mathbb{E}[|X|]$ (which we define in the obvious way) is $< \infty$, we say X is integrable. When $\mathbb{E}[X]$ is well-defined, we have

$$\mathbb{E}[X] = \sum_{x \in \Omega_X} x \cdot \mathbb{P}(X = x)$$

We have some properties of expectations as follows:

- If $X \geq 0$, then $\mathbb{E}[X] \geq 0$.
- If $X \geq 0$ and $\mathbb{E}[X] = 0$, then $\mathbb{P}(X = 0) = 1$.
- If $c \in \mathbb{R}$, then $\mathbb{E}[cX] = c\mathbb{E}[X]$ and $\mathbb{E}[c + X] = c + \mathbb{E}[X]$.
- If X and Y are random variables (and X, Y integrable), then $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$.
- In the countable case, we take $c_1, \dots, c_n \in \mathbb{R}$ and $X_1, \dots, X_n$ integrable, and then we have

$$\mathbb{E}\left[\sum_{i=1}^n c_i X_i\right] = \sum_{i=1}^n c_i \mathbb{E}[X_i]$$

Let us give a proof of the final identity in the case where Ω is countable. We have

$$\begin{aligned} \mathbb{E}\left[\sum_n X_n\right] &= \sum_{\omega \in \Omega} \left(\sum_n X_n(\omega)\right) \mathbb{P}(\{\omega\}) \\ &= \sum_n \sum_{\omega \in \Omega} X_n(\omega) \mathbb{P}(\{\omega\}) \\ &= \sum_n \mathbb{E}[X_n] \end{aligned}$$

where we can interchange sums by the assumption that the random variables are integrable.

We also have the useful result that, if $X = 1(A)$ for some event A ,

$$\mathbb{E}[X] = \mathbb{P}(A)$$

Proposition 4.13

Let $g : \mathbb{R} \rightarrow \mathbb{R}$, and define the random variable $g(X)$ by

$$g(X)(\omega) = g(X(\omega))$$

Then

$$\mathbb{E}[g(X)] = \sum_{x \in \Omega_X} g(x) \mathbb{P}(X = x)$$

Proof. Put $Y = g(X)$, then

$$\begin{aligned}\{Y = y\} &= \{\omega : Y(\omega) = y\} \\ &= \{\omega : g(X(\omega)) = y\} \\ &= \{\omega : X(\omega) \in g^{-1}(\{y\})\} \\ &= \{X \in g^{-1}(\{y\})\}\end{aligned}$$

Hence

$$\begin{aligned}\mathbb{E}[Y] &= \sum_{y \in \Omega_Y} y \cdot \mathbb{P}(Y = y) \\ &= \sum_{y \in \Omega_Y} y \cdot \mathbb{P}(X \in g^{-1}(\{y\})) \\ &= \sum_{y \in \Omega_Y} y \sum_{x \in g^{-1}(\{y\})} \mathbb{P}(X = x) \\ &= \sum_{y \in \Omega_Y} \sum_{x \in g^{-1}(\{y\})} g(x) \mathbb{P}(X = x) \\ &= \sum_{x \in \Omega_X} g(x) \cdot \mathbb{P}(X = x)\end{aligned}$$

■

Proposition 4.14

Suppose that $X \geq 0$ and takes integer values. Then

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k) = \sum_{k=0}^{\infty} \mathbb{P}(X > k)$$

Proof. Since $X \geq 0$ and $X \in \mathbb{N}$, we have

$$X = \sum_{k=1}^{\infty} 1(X \geq k) = \sum_{k=0}^{\infty} 1(X > k)$$

Taking expectations, we obtain

$$\begin{aligned}\mathbb{E}[X] &= \mathbb{E}\left[\sum_{k=1}^{\infty} 1(X \geq k)\right] &= \mathbb{E}\left[\sum_{k=0}^{\infty} 1(X > k)\right] \\ &= \sum_{k=1}^{\infty} \mathbb{E}[1(X \geq k)] &= \sum_{k=0}^{\infty} \mathbb{E}[1(X > k)] \\ &= \sum_{k=1}^{\infty} \mathbb{P}(X \geq k) &= \sum_{k=0}^{\infty} \mathbb{P}(X > k)\end{aligned}$$

■

We present another proof of the inclusion-exclusion formula. First, we should establish some properties about indicator functions:

- $1(A^c) = 1 - 1(A)$;
- $1(A \cap B) = 1(A) \cdot 1(B)$;
- $1(A \cup B) = 1 - (1 - 1(A))(1 - 1(B))$.

More generally,

$$\begin{aligned} 1(A_1 \cup \dots \cup A_n) &= 1 - \prod_{i=1}^n (1 - 1(A_i)) \\ &= \sum_{i=1}^n 1(A_i) - \sum_{1 \leq i_1 < i_2 \leq n} 1(A_{i_1} \cap A_{i_2}) + \dots + (-1)^{n+1} 1(A_1 \cap \dots \cap A_n) \end{aligned}$$

Then, we can take expectations of both sides and apply $\mathbb{E}[1(A)] = \mathbb{P}(A)$ to get

$$\mathbb{P}(A_1 \cup \dots \cup A_n) = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k})$$

§4.4 Moments, variance and covariance

Definition 4.15. Let X be a random variable and $r \in \mathbb{N}$. We call $\mathbb{E}[X^r]$, as long as it is well-defined, the r^{th} moment of X .

Definition 4.16. The variance of X is given by

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

Furthermore, the standard deviation of X is given by $\sqrt{\text{Var}(X)}$.

We have some properties of variance as follows:

- $\text{Var}(X) \geq 0$, since it is the expectation of a squared quantity;
- If $\text{Var}(X) = 0$, then $\mathbb{P}(X = \mathbb{E}[X]) = 1$;
- If $c \in \mathbb{R}$, then $\text{Var}(cX) = c^2 \text{Var}(X)$ and $\text{Var}(X + c) = \text{Var}(X)$.

Lemma 4.17

The variance of X is also given by

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

Proof. We have

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[X^2 - 2X\mathbb{E}[X] + (\mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + (\mathbb{E}[X])^2 \\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \end{aligned}$$

■

Lemma 4.18

The function $\mathbb{E}[(X - c)^2]$ is minimised when $c = \mathbb{E}[X]$, and at that point is equal to $\text{Var}(X)$.

Proof. We have

$$\begin{aligned} f(c) &= \mathbb{E}[(X - c)^2] \\ &= \mathbb{E}[X^2] - 2c\mathbb{E}[X] + c^2 \\ \implies f'(c) &= -2\mathbb{E}[X] + 2c = 0 \\ \implies c &= \mathbb{E}[X] \end{aligned}$$

■

Example 4.19 ① Suppose $X \sim \text{Bin}(n, p)$. Recall $\mathbb{E}[X] = np$. We have

$$\text{Var}(X) = \mathbb{E}[X(X - 1)] + \mathbb{E}[X] - \mathbb{E}[X]^2$$

Then

$$\begin{aligned} \mathbb{E}[X(X - 1)] &= \sum_{k=2}^n k(k-1) \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} \\ &= (n-1)np^2 \sum_{k=0}^{n-2} \binom{n-2}{k} p^k (1-p)^{n-2-k} \\ &= (n-1)np^2 \end{aligned}$$

so that $\text{Var}(X) = (n-1)np^2 + np - n^2p^2 = np(1-p)$.

② Suppose $X \sim \text{Poi}(\lambda)$. Recall $\mathbb{E}[X] = \lambda$. We have

$$\begin{aligned} \mathbb{E}[X(X - 1)] &= \sum_{k=2}^{\infty} k(k-1) e^{-\lambda} \frac{\lambda^k}{k!} \\ &= e^{-\lambda} \lambda^2 \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2 \end{aligned}$$

So $\text{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda$.

Definition 4.20. Let X and Y be random variables. Then the *covariance* of X and Y is

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

We have some properties:

- $\text{Cov}(X, Y) = \text{Cov}(Y, X)$;
- $\text{Cov}(X, X) = \text{Var}(X)$;
- for any $c \in \mathbb{R}$, $\text{Cov}(cX, Y) = c \text{Cov}(X, Y)$ and $\text{Cov}(X + c, Y) = \text{Cov}(X, Y)$;
- for any $c \in \mathbb{R}$, $\text{Cov}(c, X) = 0$.

Lemma 4.21

The covariance of X and Y is given by

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Proof. Expand. ■

Lemma 4.22

For random variables X and Y , we have

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$$

Proof. We have

$$\begin{aligned} \text{Var}(X + Y) &= \mathbb{E}[(X - \mathbb{E}[X]) + (Y - \mathbb{E}[Y])]^2 \\ &= \text{Var}(X) + 2 \text{Cov}(X, Y) + \text{Var}(Y) \end{aligned}$$
■

Lemma 4.23

For $c_1, \dots, c_n$ and $d_1, \dots, d_n \in \mathbb{R}$,

$$\text{Cov}\left(\sum_{i=1}^n c_i X_i, \sum_{j=1}^n d_j Y_j\right) = \sum_{i,j=1}^n c_i d_j \text{Cov}(X_i, Y_j)$$

Corollary 4.24

We have

$$\begin{aligned} \text{Var}\left(\sum_{i=1}^n X_i\right) &= \text{Cov}\left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i\right) \\ &= \sum_{i=1}^n \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j) \end{aligned}$$

Let us consider some results on independence.

Lemma 4.25

Suppose X_1 , X_2 and X_3 are independent – then X_1 and X_2 are independent.

Proof. We have

$$\mathbb{P}(X_1 = x_1, X_2 = x_2) = \sum_{x_3} \mathbb{P}(X_1 = x_1, X_2 = x_2, X_3 = x_3)$$

$$\begin{aligned}
&= \sum_{x_3} \mathbb{P}(X_1 = x_1) \mathbb{P}(X_2 = x_2) \mathbb{P}(X_3 = x_3) \\
&= \mathbb{P}(X_1 = x_1) \mathbb{P}(X_2 = x_2)
\end{aligned}$$

■

Remark. The result obviously generalises to any subset of a set of independent events being independent.

Lemma 4.26

Let X and Y be two independent random variables, and let $f, g : \mathbb{R} \rightarrow \mathbb{R}_+$. Then

$$\mathbb{E}[f(X)g(Y)] = \mathbb{E}[f(X)]\mathbb{E}[g(Y)]$$

Proof. Let the random variable $Z = (X, Y)$, so $h(Z) = f(X)g(Y)$. Then

$$\begin{aligned}
\mathbb{E}[h(Z)] &= \sum_{(x,y)} h(x,y) \mathbb{P}(Z = (x,y)) \\
&= \sum_{(x,y)} f(x)g(y) \mathbb{P}(X = x, Y = y) \\
&= \sum_{(x,y)} f(x)g(y) \mathbb{P}(X = x) \mathbb{P}(Y = y) \\
&= \mathbb{E}[f(X)] \cdot \mathbb{E}[g(Y)]
\end{aligned}$$

■

Proposition 4.27

If X and Y are independent, $\text{Cov}(X, Y) = 0$.

Remark. The converse is not true.

Example 4.28

Let X_1, X_2, X_3 be independent Bernoulli random variables with $p = \frac{1}{2}$. Let $Y_1 = 2X_1 - 1$, $Y_2 = 2X_2 - 1$. Also let $Z_1 = Y_1X_3$, $Z_2 = Y_2X_3$. Then $\mathbb{E}[Y_1] = \mathbb{E}[Y_2] = 0$, and $\mathbb{E}[Z_1] = \mathbb{E}[Z_2] = 0$ by independence. Then

$$\text{Cov}(Z_1, Z_2) = \mathbb{E}[Z_1Z_2] = \mathbb{E}[Y_1Y_2X_3^2]$$

But we claim Z_1 and Z_2 are not independent. Observe $\mathbb{P}(Z_1 = 0, Z_2 = 0) = \mathbb{P}(X_3 = 0) = \frac{1}{2}$. But $\mathbb{P}(Z_1 = 0) = \mathbb{P}(Z_2 = 0) = \frac{1}{2}$.

§4.5 Inequalities on random variables

Proposition 4.29 (Markov's inequality)

Let X be a non-negative random variable and $a > 0$. Then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Proof. We have

$$\begin{aligned} X &\geq a \cdot 1(X \geq a) \\ \implies \mathbb{E}[X] &\geq a \cdot \mathbb{P}(X \geq a) \end{aligned}$$

by taking expectations. ■

Proposition 4.30 (Chebyshev's inequality)

Let X be a random variable with $\mathbb{E}[X] < \infty$. Then, for any $a > 0$,

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq a) \leq \frac{\text{Var}(X)}{a^2}$$

Proof. Observe

$$\begin{aligned} \mathbb{P}(|X - \mathbb{E}[X]| \geq a) &= \mathbb{P}((X - \mathbb{E}[X])^2 \geq a^2) \\ &\leq \frac{\mathbb{E}[(X - \mathbb{E}[X])^2]}{a^2} && \text{by Markov} \\ &= \frac{\text{Var}(X)}{a^2} \end{aligned}$$
■

Proposition 4.31 (Cauchy-Schwarz)

Let X and Y be random variables, then

$$\mathbb{E}[|XY|] \leq \sqrt{\mathbb{E}[X^2] \cdot \mathbb{E}[Y^2]}$$

Proof. Assume that $\mathbb{E}[X^2]$ and $\mathbb{E}[Y^2]$ are finite, otherwise there is nothing to prove. Then, $|XY| \leq \frac{1}{2}(X^2 + Y^2)$, so, taking expectations, $\mathbb{E}[|XY|]$ is also finite. We may also assume $\mathbb{E}[X^2] > 0$, $\mathbb{E}[Y^2] > 0$. Let us also assume for the time being that $X, Y \geq 0$.

Let $t \in \mathbb{R}$ and consider $(X - tY)^2$. We have

$$\begin{aligned} 0 &\leq \mathbb{E}[(X - tY)^2] \\ &= \mathbb{E}[X^2] - 2t\mathbb{E}[XY] + t^2\mathbb{E}[Y^2] := f(t) \end{aligned}$$

Then $f'(t) = -2\mathbb{E}[XY] + 2t\mathbb{E}[Y^2]$, so $\min f(t)$ is achieved at $t^* = \frac{\mathbb{E}[XY]}{\mathbb{E}[Y^2]}$. We hence have

$$0 \leq \mathbb{E}[X^2] - 2 \frac{\mathbb{E}[XY]^2}{\mathbb{E}[Y^2]} + \frac{\mathbb{E}[XY]^2}{\mathbb{E}[Y^2]}$$

$$\implies \mathbb{E}[XY]^2 \leq \mathbb{E}[X^2] \cdot \mathbb{E}[Y^2]$$

■

Remark. We could also have just put in our value for t^* from the beginning – the minimisation approach is purely to provide rationale.

Remark. When is equality achieved? We would need $f(t^*) = 0$, that is

$$\begin{aligned} \mathbb{E}[(X - t^*Y)^2] &= 0 \\ \implies P(X = t^*Y) &= \mathbb{P}(X - t^*Y = 0) = 1 \end{aligned}$$

so X is just a multiple of Y .

Definition 4.32 (Convexity). A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called *convex* if, for any $x, y \in \mathbb{R}$ and any $t \in [0, 1]$,

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y)$$

Proposition 4.33 (Jensen's inequality)

Let X be a random variable, and $f : \mathbb{R} \rightarrow \mathbb{R}$ a convex function. Then

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$$

Remark. We can remember the direction of the inequality by putting $f(x) = x^2$ and using non-negativity of the variance.

Lemma 4.34

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function. Then f is the supremum of all the lines lying below it. In other words, for all m in $\mathbb{R}$, there exist a, b in $\mathbb{R}$ such that $f(m) = am + b$ and $f(x) \geq ax + b$ for all x .

Proof of lemma. Let $m \in \mathbb{R}$, and suppose $x < m < y$. Then $m = tx + (1 - t)y$ for some $t \in [0, 1]$, or, equivalently,

$$t(m - x) = (1 - t)(y - m) \tag{*}$$

By convexity, we have

$$\begin{aligned} f(m) &= f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y) \\ \implies t(f(m) - f(x)) &\leq (1 - t)(f(y) - f(m)) \\ (*) \implies \frac{f(m) - f(x)}{m - x} &\leq \frac{f(y) - f(m)}{y - m} \end{aligned}$$

Then set

$$a = \sup_{x < m} \frac{f(m) - f(x)}{m - x}$$

so that, for any $x < m < y$,

$$\frac{f(m) - f(x)}{m - x} \leq a \leq \frac{f(y) - f(m)}{y - m}$$

Hence, by rearrangement, we conclude $f(z) \geq a(z - m) + f(m)$, for any z (depending on whether z is greater than m or less than m , we use either the left or right inequality.) ■

Proof of proposition. Let $m = \mathbb{E}[X]$. By the claim, we have $a, b \in \mathbb{R}$ such that $f(m) = am + b$ and $f(x) \geq ax + b$ for all x . We have

$$\begin{aligned} f(X) &\geq aX + b \\ \implies \mathbb{E}[f(X)] &\geq a\mathbb{E}[X] + b = am + b = f(m) = f(\mathbb{E}[X]) \end{aligned}$$

■

Remark. When is equality achieved?

Suppose f convex with the extra property that, for $n = \mathbb{E}[X]$, there exist $a, b \in \mathbb{R}$ with $f(m) = am + b$ but $f(x) > ax + b$, note the strict inequality, for $x \neq m$. When should X satisfy $\mathbb{E}[f(X)] = f(\mathbb{E}[X])$?

Since $f(X) \geq aX + b$, we have $f(X) - (aX + b) \geq 0$. Taking expectations, we have

$$\mathbb{E}[f(X)] \geq a\mathbb{E}[X] + b = am + b = f(m) = f(\mathbb{E}[X])$$

Since we want $\mathbb{E}[f(X)] = f(\mathbb{E}[X])$, we must have

$$\begin{aligned} \mathbb{E}[f(X) - (aX + b)] &= 0 \\ \implies \mathbb{P}(f(X) = aX + b) &= 1 \end{aligned}$$

but then, since $f(x) > ax + b$ for $x \neq m$, we conclude $\mathbb{P}(X = m) = 1$ (if we didn't have that extra part, we'd conclude f is just a linear function).

Lecture $n + 1$

Proposition 4.35 (AM-GM inequality)

Let $x_1, \dots, x_n > 0$. Then

$$\left(\prod_{k=1}^n x_k \right)^{\frac{1}{n}} \leq \frac{1}{n} \sum_{k=1}^n x_k$$

Lemma 4.36

Let f be a convex function. Then, for any $x_1, \dots, x_n \in \mathbb{R}$ we have

$$\frac{1}{n} \sum_{k=1}^n f(x_k) \geq f\left(\frac{1}{n} \sum_{k=1}^n x_k\right)$$

Proof of lemma. Let X be a random variable with $\mathbb{P}(X = x_i) = \frac{1}{n}$, $i = 1, \dots, n$. Then

$$\frac{1}{n} \sum_{k=1}^n f(x_k) = \sum_x f(x) \mathbb{P}(X = x) = \mathbb{E}[f(X)]$$

so Jensen's gives

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]) = f\left(\frac{1}{n} \sum_{k=1}^n x_k\right)$$

as required. ■

Proof of proposition. Let $f(x) = -\log x$ and suppose $x_1, \dots, x_n > 0$. Then

$$\begin{aligned} -\frac{1}{n} \sum_{k=1}^n \log x_k &\geq -\log \left(\frac{1}{n} \sum_{k=1}^n x_k \right) \\ \implies \left(\prod_{k=1}^n x_k \right)^{\frac{1}{n}} &\leq \frac{1}{n} \sum_{k=1}^n x_k \end{aligned}$$

as required. ■

§4.6 Joint distribution and conditional expectation

Recall if X is a discrete random variable, then the distribution of X is $(\mathbb{P}(X = x))_x$.

Definition 4.37. Let $X_1, \dots, X_n$ be discrete random variables. Their *joint distribution* is defined to be the set of values of

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n)$$

for any $x_1, \dots, x_n$.

Remark. We can recover the distribution of any one of the variables from such a joint distribution with

$$\mathbb{P}(X_1 = x_1) = \sum_{x_2, \dots, x_n} \mathbb{P}(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

In this context the distribution of the single variable is sometimes called its *marginal* distribution.

Definition 4.38. Let X and Y be discrete random variables. The *conditional distribution* of X given $Y = y$ is

$$\mathbb{P}(X = x | Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}$$

By the law of probability we of course have

$$\mathbb{P}(X = x) = \sum_y \mathbb{P}(X = x | Y = y) \mathbb{P}(Y = y)$$

Let X and Y be independent – what is the distribution of $X + Y$? We have

$$\begin{aligned}\mathbb{P}(X + Y = z) &= \sum_y \mathbb{P}(X + Y = z, Y = y) \\ &= \sum_y \mathbb{P}(X = z - y, Y = y) \\ &= \sum_y \mathbb{P}(X = z - y) \mathbb{P}(Y = y)\end{aligned}$$

where the last line is by independence. This is of course a convolution.

Example

Let $X \sim \text{Poi}(\lambda)$ and $Y \sim \text{Poi}(\mu)$, with $\lambda, \mu > 0$ and $X \perp Y$. Then

$$\begin{aligned}\mathbb{P}(X + Y = n) &= \sum_{k=0}^n \mathbb{P}(X = n - k) \mathbb{P}(Y = k) \\ &= e^{-\lambda} \frac{\lambda^{n-k}}{(n-k)!} \cdot e^{-\mu} \frac{\mu^k}{k!} \\ &= \frac{e^{-(\lambda+\mu)}}{n!} \sum_{k=0}^n \binom{n}{k} \lambda^{n-k} \mu^k \\ &= e^{-(\lambda+\mu)} \frac{(\lambda + \mu)^n}{n!}\end{aligned}$$

so in particular $X + Y \sim \text{Poi}(\lambda + \mu)$.

Let us derive conditional expectation for random variables. First, take $B \in \mathcal{F}$ to be an event with $\mathbb{P}(B) > 0$. We have

$$\mathbb{E}[X|B] = \frac{\mathbb{E}[X \cdot 1(B)]}{\mathbb{P}(B)}$$

and, if $X = 1(A)$, then

$$\mathbb{E}[1(A)|B] = \frac{\mathbb{E}[1(A) \cdot 1(B)]}{\mathbb{P}(B)} = \frac{\mathbb{E}[1(A \cap B)]}{\mathbb{P}(B)} = \mathbb{P}(A|B)$$

We can now formulate the *law of total expectation*.

Proposition 4.39 (Law of total expectation)

Let X be a random variable and (Ω_n) a partition of Ω with $\mathbb{P}(\Omega_n) > 0$ for any n . Then

$$\mathbb{E}[X] = \sum_n \mathbb{E}[X|\Omega_n] \mathbb{P}(\Omega_n)$$

Proof. Use indicators. We have

$$\mathbb{E}[X] = \mathbb{E}[X \cdot 1(\Omega)] = \mathbb{E}\left[X \cdot \sum_n 1(\Omega_n)\right] = \sum_n \mathbb{E}[X|\Omega_n] \mathbb{P}(\Omega_n)$$

■

Now, we want to condition expectation not on an event but on a particular random variable. We have

$$\begin{aligned}\mathbb{E}[X|Y = y] &= \frac{\mathbb{E}[X \cdot 1(Y = y)]}{\mathbb{P}(Y = y)} \\ &= \sum_x x \cdot \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)} \\ &= \sum_x x \cdot \mathbb{P}(X = x|Y = y)\end{aligned}$$

Let us now define $g(y) = \mathbb{E}[X|Y = y]$.

Definition 4.40. The expectation of X given Y , $\mathbb{E}[X|Y]$, is a random variable which is a function of Y . In particular,

$$\mathbb{E}[X|Y] = g(Y)$$

Example

Toss a p -coin independently n times, and let $X_i = 1(i^{\text{th}} \text{ toss is H})$. Let $Y_n = X_1 + \dots + X_n$. Let us attempt to calculate $\mathbb{E}[X_1|Y_n]$. First, let

$$\begin{aligned}g(y) = \mathbb{E}[X_1|Y_n = y] &= \frac{\mathbb{E}[X_1 \cdot 1(Y_n = y)]}{\mathbb{P}(Y_n = y)} \\ &= \frac{\mathbb{P}(X_1 = 1, Y_n = y)}{\mathbb{P}(Y_n = y)} \\ &= \frac{p \binom{n-1}{y-1} p^{y-1} (1-p)^{n-y}}{\binom{n}{y} p^y (1-p)^{n-y}} \\ &= \frac{y}{n}\end{aligned}$$

Hence, $\mathbb{E}[X_1|Y_n] = \frac{Y_n}{n}$.

We have some simple and uninspiring properties:

- $\mathbb{E}[cX|Y] = c\mathbb{E}[X|Y]$;
- $\mathbb{E}[c + X|Y] = c + \mathbb{E}[X|Y]$;
- $\mathbb{E}[c|Y] = c$;
- If $X_1, \dots, X_n$ are random variables,

$$\mathbb{E}\left[\sum_{i=1}^n X_i|Y\right] = \sum_{i=1}^n \mathbb{E}[X_i|Y]$$

and some more interesting properties. We have

$$\begin{aligned}\mathbb{E}[X|Y] &= \sum_y 1(Y = y) \mathbb{E}[X|Y = y] \\ \implies \mathbb{E}[\mathbb{E}[X|Y]] &= \sum_y \mathbb{E}[X|Y = y] \mathbb{P}(Y = y) = \mathbb{E}[X]\end{aligned}$$

Suppose now X and Y are independent random variables. Then

$$\begin{aligned}
 \mathbb{E}[X|Y] &= \sum_y 1(Y = y) \mathbb{E}[X|Y = y] \\
 &= \sum_y 1(Y = y) \sum_x x \cdot \mathbb{P}(X = x|Y = y) \\
 &= \sum_y 1(Y = y) \sum_x x \cdot \mathbb{P}(X = x) \\
 &= \sum_y 1(Y = y) \mathbb{E}[X] = \mathbb{E}[X]
 \end{aligned}$$

Proposition 4.41

Let Y and Z be independent. Then

$$\mathbb{E}[\mathbb{E}[X|Y]|Z] = \mathbb{E}[X]$$

Proof. Let $g(Y) = \mathbb{E}[X|Y]$, $g(y) = \mathbb{E}[X|Y = y]$. We claim $g(Y)$ is independent of Z . Observe

$$\begin{aligned}
 \mathbb{P}(g(Y) = w, Z = z) &= \sum_{y \in g^{-1}(\{w\})} \mathbb{P}(Y = y, Z = z) \\
 &= \sum_{y \in g^{-1}(\{w\})} \mathbb{P}(Y = y) \mathbb{P}(Z = z) \\
 &= \mathbb{P}(g(Y) = w) \mathbb{P}(Z = z)
 \end{aligned}$$

Hence, previous results give $\mathbb{E}[g(Y)|Z] = \mathbb{E}[g(Y)]$, and this is $\mathbb{E}[\mathbb{E}[X|Y]] = \mathbb{E}[X]$. ■

Proposition 4.42

Let X, Y be two random variables, and $h : \mathbb{R} \rightarrow \mathbb{R}$. Then

$$\mathbb{E}[h(Y) \cdot X|Y] = h(Y) \cdot \mathbb{E}[X|Y]$$

Proof. We have

$$\begin{aligned}
 \mathbb{E}[h(Y) \cdot X|Y] &= \sum_y 1(Y = y) \mathbb{E}[h(Y) \cdot X|Y = y] \\
 &= \sum_y 1(Y = y) h(y) \mathbb{E}[X|Y = y] \\
 &= h(Y) \sum_y 1(Y = y) \mathbb{E}[X|Y = y] \\
 &= h(Y) \mathbb{E}[X|Y]
 \end{aligned}$$
■

Corollary 4.43

$$\mathbb{E}[\mathbb{E}[X|Y]|Y] = \mathbb{E}[X|Y]$$

Example

Toss a p -coin independently n times, and let $X_i = 1$ (i^{th} toss is H). Let $Y_n = X_1 + \dots + X_n$. What is $\mathbb{E}[X_1|Y_n]$?

By symmetry, $\mathbb{E}[X_1|Y_n] = \mathbb{E}[X_i|Y_n]$ for any i , $1 \leq i \leq n$. Then

$$Y_n = \mathbb{E}[Y_n|Y_n] = \mathbb{E}[X_1 + \dots + X_n|Y_n] = n\mathbb{E}[X_1|Y_n]$$

which was what we wanted.

§5 Random walks

Definition 5.1 (Random process). A *random process* or *stochastic process* is a sequence of random variables (X_n) .

Definition 5.2 (Random walks). A *random walk* is a random process that can be expressed as

$$X_n = x + Y_1 + \dots + Y_n$$

where x is a number and (Y_i) are independent identically distributed (i.i.d.) random variables. We call the Y_i the *steps* of the random variable.

A *simple random walk* has

$$Y_i = \begin{cases} 1 & \text{with probability } p \\ -1 & \text{with probability } q = 1 - p \end{cases}$$

We will write

- $\mathbb{P}_x(A) = \mathbb{P}(A|X_0 = x)$;
- $h(x) = \mathbb{P}_x((X_n) \text{ reaches } a \text{ before } 0)$, where a is some arbitrary constant at which the game ends.

so $h(0) = 0$, $h(a) = 1$. Then

$$\begin{aligned} h(x) &= \mathbb{P}_x((X_n) \text{ hits } a \text{ bef. } 0) \\ &= \mathbb{P}_x((X_n) \text{ hits } a \text{ bef. } 0 | Y_1 = 1) \cdot p + \mathbb{P}_x((X_n) \text{ hits } a \text{ bef. } 0 | Y_1 = -1) \cdot q \\ &= p \cdot h(x+1) + q \cdot h(x-1) \end{aligned}$$

- Case $p = q = \frac{1}{2}$, then X_n is a *simple symmetric random walk*. We have

$$\begin{aligned} h(x) &= \frac{1}{2}h(x+1) + \frac{1}{2}h(x-1) \\ \implies h(x+1) - h(x) &= h(x) - h(x-1) \\ \implies \boxed{h(x) = cx} \end{aligned}$$

for some constant c . Since $h(a) = 1$, $c = \frac{1}{a}$, and hence $h(x) = \frac{x}{a}$.

- Case $p \neq q$. Let us try a solution of the form λ^x for some λ – we have

$$\begin{aligned} \lambda^x &= p\lambda^{x+1} + q\lambda^{x-1} \\ \implies \lambda &= 1, \lambda = \frac{q}{p} \end{aligned}$$

so the general solution has the form

$$h(x) = A + B \left(\frac{q}{p}\right)^x$$

which, applying boundary conditions, gives

$$\boxed{h(x) = \frac{\left(\frac{q}{p}\right)^x - 1}{\left(\frac{q}{p}\right)^a - 1}}$$

These results are called the *gambler's ruin estimates*.

Now let us consider a new quantity – let $T = \min\{n \geq 0 : X_n \in \{0, a\}\}$ – the first n such that X_n is either 0 or a , i.e. the duration of the game. We want to find $\mathbb{E}[T|X_0 = x] = \mathbb{E}_x[T] = k(x)$. We have from the law of total expectation

$$\begin{aligned} k(x) &= \mathbb{E}_x[T|Y_1 = 1] \cdot \mathbb{P}(Y_1 = 1) + \mathbb{E}_x[T|Y_1 = -1] \cdot \mathbb{P}(Y_1 = -1) \\ &= p(1 + k(x+1)) + q(1 + k(x-1)) \\ &= 1 + pk(x+1) + qk(x-1) \end{aligned}$$

with $k(0) = k(a) = 0$.

- Case $p = q = \frac{1}{2}$. Try a solution of the form Ax^2 , to get

$$\begin{aligned} Ax^2 &= 1 + pA(x+1)^2 + qA(x-1)^2 \\ \implies A &= -1 \end{aligned}$$

giving a general solution of the form $k(x) = -x^2 + Bx + C$. Applying boundary conditions, we obtain

$$\boxed{k(x) = x(a-x)}$$

- Case $p \neq q$. Try $k(x) = Cx$ to get

$$\begin{aligned} Cx &= 1 + pC(x+1) + qC(x-1) \\ \implies c &= \frac{1}{q-p} \end{aligned}$$

Adding this to the solution of the homogeneous equation, we obtain

$$k(x) = A + \frac{1}{q-p}x + B\left(\frac{q}{p}\right)^x$$

which, after boundary conditions, gives

$$k(x) = \frac{1}{q-p}x - \frac{a}{q-p} \cdot \frac{\left(\frac{q}{p}\right)^x - 1}{\left(\frac{q}{p}\right)^a - 1}$$

§6 Probability generating functions

Let X be a random variable with values in $\mathbb{N}$ – recall $p_r = \mathbb{P}(X = r)$ for $r \in \mathbb{N}$ constitutes the probability mass function.

Definition 6.1 (Probability generating function). The *probability generating function* (PGF) for a random variable X with values in $\mathbb{N}$ and probability masses (p_r) is

$$p(z) = \mathbb{E}[z^X] = \sum_{r=0}^{\infty} p_r z^r$$

with $|z| \leq 1$.

Remark. This is well-defined because

$$|p(z)| = \left| \sum_{r=0}^{\infty} p_r z^r \right| \leq \sum_{r=0}^{\infty} p_r |z|^r \leq \sum_{r=0}^{\infty} p_r = 1$$

so p converges absolutely for $|z| \leq 1$.

Theorem 6.2

The pgf uniquely determines the distribution of X .

Proof. Let (p_r) and (q_r) be two probability distributions with the same pgf's – for all $|z| \leq 1$, $\sum p_r z^r = \sum q_r z^r$.

Plugging in $z = 0$ gives $p_0 = q_0$. We now proceed by strong induction. Suppose $p_r = q_r$ for all $r \leq n$, so

$$\sum_{r=n+1}^{\infty} p_r z^r = \sum_{r=n+1}^{\infty} q_r z^r$$

Clearly dividing by z^{n+1} and setting $z = 0$ gives $p_{n+1} = q_{n+1}$ as required. ■

Proposition 6.3

Suppose $X_1, \dots, X_n$ are independent random variables with pgf's $q_i(z)$, and let $X = X_1 + \dots + X_n$ have pgf p . Then

$$p(z) = q_1(z) \dots q_n(z)$$

Proof. Clearly

$$p(z) = \mathbb{E}[z^{X_1 + \dots + X_n}] = \mathbb{E}[X_1] \dots \mathbb{E}[X_n] = q_1(z) \dots q_n(z)$$

by independence. ■

Example

Let us calculate some pgf's for important distributions.

① *Binomial.* Suppose $X \sim \text{Bin}(n, p)$. Then

$$\begin{aligned} p(z) &= \sum_{k=0}^n z^k \binom{n}{k} p^k (1-p)^{n-k} \\ &= (pz + 1 - p)^n \end{aligned}$$

② *Geometric.* Suppose $X \sim \text{Geo}(p)$. Then

$$\begin{aligned} \mathbb{E}[z^X] &= \sum_{r=0}^{\infty} z^r (1-p)^r p \\ &= \frac{p}{1 - z(1-p)} \end{aligned}$$

③ *Poisson.* Suppose $X \sim \text{Poi}(\lambda)$. Then

$$\begin{aligned} \mathbb{E}[z^X] &= \sum_{r=0}^{\infty} z^r e^{-\lambda} \frac{\lambda^r}{r!} \\ &= e^{\lambda(z-1)} \end{aligned}$$

Example

This fact can make calculating the pmf of sums of random variables easier. For example, suppose $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$ with $X \perp Y$. Then

$$\begin{aligned} \mathbb{E}[z^{X+Y}] &= \mathbb{E}[z^X] \mathbb{E}[z^Y] = (pz + 1 - p)^n \cdot (pz + 1 - p)^m \\ &= (pz + 1 - p)^{n+m} \end{aligned}$$

from which it follows that $X + Y \sim \text{Bin}(n + m, p)$.

What about Poisson? Suppose $X \sim \text{Poi}(\lambda)$, $Y \sim \text{Poi}(\mu)$, and $X \perp Y$. Then

$$\mathbb{E}[z^{X+Y}] = e^{\lambda(z-1)} e^{\mu(z-1)} = e^{(\lambda+\mu)(z-1)}$$

from which it follows that $X + Y \sim \text{Poi}(\lambda + \mu)$.

For the following results, it will be helpful to introduce the notation

$$p(1^-) = \lim_{z \rightarrow 1^-} p(z)$$

Proposition 6.4

Let X be a discrete random variable with pgf $p(z)$, then

$$\mathbb{E}[X] = p'(1^-)$$

Proof. • Assume $\mathbb{E}[X] < \infty$, and take $0 < z < 1$. We have

$$p'(z) = \sum_{r=1}^{\infty} r p_r z^{r-1} \leq \sum_{r=1}^{\infty} r p_r = \mathbb{E}[X]$$

Hence p' is upper bounded by $\mathbb{E}[X]$, and it is clearly increasing, so the limit exists and is $\leq \mathbb{E}[X]$.

Let $\varepsilon > 0$. There exists N such that

$$\sum_{r=1}^N r p_r \geq \mathbb{E}[X] - \varepsilon$$

since this is what it means for the infinite sum to be equal to $\mathbb{E}[X]$. Furthermore,

$$\lim_{z \rightarrow 1^-} p'(z) \geq \lim_{z \rightarrow 1^-} \sum_{r=1}^N r p_r z^{r-1} = \sum_{r=1}^N r p_r \geq \mathbb{E}[X] - \varepsilon$$

• If $\mathbb{E}[X] = \infty$, then, for any $M \geq 0$, we have N such that

$$\sum_{r=1}^N r p_r \geq M$$

Then, as above,

$$\lim_{z \rightarrow 1^-} p'(z) \geq \lim_{z \rightarrow 1^-} \sum_{r=1}^N r p_r z^{r-1} = \sum_{r=1}^N r p_r \geq M$$

so the limit is infinite. ■

Proposition 6.5

Let X be a discrete random variable with pgf $p(z)$, then

$$\mathbb{E}[X(X-1)\dots(X-k+1)] = p^{(k)}(1^-)$$

Corollary 6.6

Let X be a discrete random variable with pgf $p(z)$, then

$$\text{Var}(X) = p''(1^-) + p'(1^-) - (p'(1^-))^2$$

Proposition 6.7

Let X be a discrete random variable with pgf $p(z)$, then

$$\mathbb{P}(X = n) = \frac{1}{n!} \left. \frac{d^n p}{dz^n} \right|_{z=0}$$

Remark. This is another way to see [theorem 6.2](#).

§6.1 Sum of a random number of random variables

Let $X_1, X_2, \dots$ be i.i.d. random variables and let N be an independent random variable with values in $\mathbb{N}$. Set

$$S_n = X_1 + \dots + X_n$$

that, is

$$S_n(\omega) = \sum_{i=1}^n X_i(\omega)$$

Passing to the random variable N , we obtain

$$S_N(\omega) = S_{N(\omega)}(\omega) = \sum_{i=1}^{N(\omega)} X_i(\omega)$$

Lemma 6.8

Let $q(z) = \mathbb{E}[z^N]$ be the pgf of N , and $p(z) = \mathbb{E}[z^{X_i}]$ be the pgf of X_i . Then

$$r(z) = \mathbb{E}[z^{S_N}] = q(p(z))$$

Proof 1. We have

$$\begin{aligned} r(z) &= \mathbb{E}[z^{S_N}] = \sum_{n=0}^{\infty} \mathbb{E}[z^{S_N} \cdot 1(N = n)] \\ &= \sum_{n=0}^{\infty} \mathbb{E}[z^{S_n} \cdot 1(N = n)] \\ &= \sum_{n=0}^{\infty} \mathbb{E}[z^{X_1 + \dots + X_n} \cdot 1(N = n)] \\ &= \sum_{n=0}^{\infty} \mathbb{E}[z^{X_1 + \dots + X_n}] \mathbb{P}(N = n) \\ &= \sum_{n=0}^{\infty} (p(z))^n \mathbb{P}(N = n) \\ &= q(p(z)) \end{aligned}$$

■

Proof 2. We have

$$r(z) = \mathbb{E}[z^{S_n}] = \mathbb{E}[\mathbb{E}[z^{S_n} | N]] = g(N)$$

where

$$\begin{aligned} g(n) &= \mathbb{E}[z^{S_n} | N = n] \\ &= \mathbb{E}[z^{S_n}] \\ &= p(z)^n \end{aligned}$$

so that

$$\begin{aligned} r(z) &= \mathbb{E}[p(z)^N] \\ &= q(p(z)) \end{aligned}$$

■

Corollary 6.9

We have

$$\begin{aligned} \mathbb{E}[S_n] &= \mathbb{E}[N] \cdot \mathbb{E}[X_1] \\ \text{Var}(S_n) &= \mathbb{E}[N] \cdot \text{Var}(X_1) + \text{Var}(N) \cdot \mathbb{E}[X_1]^2 \end{aligned}$$

Proof. The first result follows from

$$\begin{aligned} \mathbb{E}[S_n] &= r'(1-) \\ &= \lim_{z \rightarrow 1-} r'(z) \\ &= \lim_{z \rightarrow 1-} q'(p(z)) \lim_{z \rightarrow 1-} p'(z) \end{aligned}$$

Clearly $p(z)$ approaches 1 from below as z approaches 1 from below, so this is just

$$q'(1-)p'(1-) = \mathbb{E}[N] \cdot \mathbb{E}[X_1]$$

as required.

The second result is left as an exercise. ■

§7 Branching processes

Think of a branching process as a series of generations of offspring. We have $X_0 = 1$, there being one person in the first generation; then, in each generation, each individual produces k offspring independently with a fixed distribution (in particular, the distribution of X_1).

Let $Y_{k,n}$ with $k \geq 1$, $n \geq 0$ be i.i.d. random variables with the distribution of X_1 , interpreted as the number of offspring produced by the k^{th} individual in the n^{th} generation. Hence

$$X_{n+1} = \begin{cases} Y_{1,n} + \cdots + Y_{X_n,n} & X_n \geq 1 \\ 0 & X_n = 0 \end{cases}$$

Proposition 7.1

We have

$$\mathbb{E}[X_n] = \mathbb{E}[X_1]^n$$

Proof. We will proceed by induction. The base case is clear. Let us consider the conditional expectation $\mathbb{E}[X_{n+1}|X_n = m]$ – we have

$$\begin{aligned} \mathbb{E}[X_{n+1}|X_n = m] &= \mathbb{E}[Y_{1,n} + \cdots + Y_{m,n}|X_n = m] \\ &= \mathbb{E}[Y_{1,n} + \cdots + Y_{m,n}] && \text{(independence)} \\ &= m\mathbb{E}[X_1] \end{aligned}$$

so that $\mathbb{E}[X_{n+1}|X_n] = X_n\mathbb{E}[X_1]$. Hence $\mathbb{E}[X_{n+1}] = \mathbb{E}[\mathbb{E}[X_{n+1}|X_n]] = \mathbb{E}[X_n\mathbb{E}[X_1]] = \mathbb{E}[X_n]\mathbb{E}[X_1]$, which completes the result by induction. ■

Proposition 7.2

Let $G(z) = \mathbb{E}[z^{X_1}]$, $G_n(z) = \mathbb{E}[z^{X_n}]$, then

$$G_{n+1}(z) = G(G_n(z))$$

Proof. Again, let us consider $\mathbb{E}[z^{X_{n+1}}|X_n = m]$. We have

$$\begin{aligned} \mathbb{E}[z^{X_{n+1}}|X_n = m] &= \mathbb{E}[z^{Y_{1,n} + \cdots + Y_{m,n}}|X_n = m] \\ &= \mathbb{E}[z^{Y_{1,n} + \cdots + Y_{m,n}}] \\ &= (G(z))^m \end{aligned}$$

Hence $G_{n+1}(z) = \mathbb{E}[\mathbb{E}[z^{X_{n+1}}|X_n]] = \mathbb{E}[G(z)^{X_n}] = G_n(G(z))$ ■

§7.1 Extinction probability

Let $q_n = \mathbb{P}(X_n = 0)$, and $q = \mathbb{P}(X_n = 0 \text{ for some } n \geq 0)$. Letting A_n be the event that $X_n = 0$ for some n , we can see $q = \mathbb{P}(\cup A_n)$, and $A_n \subseteq A_{n+1}$, so the A_n form an increasing sequence and we get

$$\mathbb{P}(A_n) \rightarrow \mathbb{P}\left(\bigcup A_n\right)$$

by [proposition 1.8](#) (continuity of probability measures).

Lemma 7.3

Let $G(z) = \mathbb{E}[z^{X_1}]$, then $q_{n+1} = G(q_n)$ and $q = G(q)$.

Proof 1. We have

$$\begin{aligned} q_{n+1} &= \mathbb{P}(X_{n+1} = 0) \\ &= G_{n+1}(0) \end{aligned}$$

$$\begin{aligned} &= G(G_n(0)) \\ &= G(q_n) \end{aligned}$$

Then, since G is continuous and $q_n \rightarrow q$, we can simply take the limit on both sides to get $q = G(q)$ as required. ■

Proof 2. We will condition q_{n+1} on the number of individuals in the first generation, i.e. X_1 . Supposing $X_1 = m$, let $X_n^{(1)}, \dots, X_n^{(m)}$ be i.i.d. branching processes with the same distribution as the original one. Hence

$$X_{n+1} = X_n^{(1)} + \dots + X_n^{(m)}$$

So

$$\begin{aligned} q_{n+1} &= \mathbb{P}(X_{n+1} = 0) \\ &= \sum_m \mathbb{P}(X_{n+1} = 0 | X_1 = m) \mathbb{P}(X_1 = m) \\ &= \sum_m \mathbb{P}(X_n^{(1)} + \dots + X_n^{(m)} = 0 | X_1 = m) \mathbb{P}(X_1 = m) \\ &= \sum_m \left(\mathbb{P}(X_n^{(1)} = 0) \right)^m \mathbb{P}(X_1 = m) \\ &= \sum_m q_n^m \mathbb{P}(X_1 = m) \\ &= G(q_n) \end{aligned}$$

Once again, we take the limit on both sides. ■

Lecture 14 – 23/02/26

Recalling [proposition 7.1](#), we have that

- if $\mathbb{E}[X_1] < 1$, $\mathbb{E}[X_n] \rightarrow 0$ as $n \rightarrow \infty$, so we expect $q = 1$;
- if $\mathbb{E}[X_1] > 1$, $\mathbb{E}[X_n] \rightarrow \infty$ as $n \rightarrow \infty$, so we expect $q < 1$;
- if $\mathbb{E}[X_1] = 1$, then $\mathbb{E}[X_n] = 1$ for all n . But it is not clear what q will be. Let us focus on this last case.

We can also interpret these results in the context of a graph of G . We know that the graph of G will intersect x at $x = 1$, and we could either have another intersection in the range $0 \leq x < 1$, or no intersections.

- *There is another intersection.* Then iterating G starting from $G(0)$ will lead us to the other intersection. Furthermore, the gradient of G at 1, $G'(1-) = \mathbb{E}[X_1]$ must be greater than 1 for there to be such an intersection, as we expect;
- *There is not another intersection.* Then iterating G will take us all the way to 1, and the gradient of G at 1, $G'(1-) = \mathbb{E}[X_1]$ must be less than or equal to 1 for there not to be any further intersections. We conclude that $q = 1$ in the case $\mathbb{E}[X_1] = 1$.

Proposition 7.4

Assume $\mathbb{P}(X_1 = 1) < 1$. The extinction probability q is the minimal non-negative solution to $G(x) = x$. Furthermore, $q < 1$ if and only if $\mathbb{E}[X_1] > 1$.

Proof. We know $q = G(q)$. Let t be the smallest minimal solution of $x = G(x)$. We will show by induction that, for any n , $q_n \leq t$, and then take limits to establish $q \leq t \implies q = t$.

For the base case, $0 = q_0 \leq t$. Suppose $q_n \leq t$, then

$$q_{n+1} = G(q_n) \leq G(t) = t$$

since G is increasing.

Now, we show $q < 1$ if and only if $\mathbb{E}[X_1] > 1$. Suppose for now $g_0 + g_1 = \mathbb{P}(X_1 \leq 1) < 1$.

Let $H(z) = G(z) - z = \sum_{r=0}^{\infty} g_r z^r - z$, so $H(1) = 0$. First, we show that H can have at most one root in $[0, 1)$. Observe

$$H''(z) = \sum_{r=0}^{\infty} r(r-1)g_r z^{r-2} > 0$$

in $(0, 1)$, because $g_0 + g_1 < 1$, so there must be some later r with a positive value of g_r , making the sum strictly larger than zero. Hence $H'(z)$ is strictly increasing, and so can have at most one more root other than 1 – otherwise, we would have more than one point c with $H'(c) = 0$ by Rolle's theorem.

We now split into two cases.

- H has no root other than 1, so $q = 1$. Then $H'(1-) = G'(1-) - 1 = \mathbb{E}[X_1] - 1$. Furthermore, $H(0) = G(0) = g_0 \geq 0$ while $H(1) = 0$, so $H(z) \geq 0$ for all $z \in [0, 1]$. Now

$$H'(1-) = \lim_{z \rightarrow 1-} \frac{H(z) - H(1)}{z - 1} \leq 0$$

since the numerator is $H(z) - 0 \geq 0$ while the denominator is $z - 1 \leq 0$. Hence $H'(1-) \leq 0 \implies \mathbb{E}[X_1] \leq 1$, as required.

- H has one more root r , so r is the extinction probability. We hence have $H(r) = 0$ and $H(1) = 0$ with $r < 1$, so we have $z \in (r, 1)$ with $H'(z) = 0$ by Rolle. Hence $\mathbb{E}[X_1] - 1 = H'(-1) > H'(z) = 0$, so $\mathbb{E}[X_1] > 1$ as required.

On the other hand, if $\mathbb{P}(X_1 \leq 1) = 1$, then $\mathbb{E}[X_1] = g_1$, so

$$\begin{aligned} G(z) &= g_0 + g_1 z \\ &= 1 - \mathbb{E}[X_1] + \mathbb{E}[X_1]z \end{aligned}$$

Then, if $G(z) = z$, we have $(1 - \mathbb{E}[X_1])z = 1 - \mathbb{E}[X_1]$ and, since $\mathbb{E}[X_1] = g_1 < 1$, $1 - \mathbb{E}[X_1] \neq 0$ so $z = 1$. ■

§8 Continuous random variables

Recall [definition 4.2](#), that

Definition. Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a *random variable* is a function

$$X : \Omega \rightarrow \mathbb{R}$$

satisfying

$$\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}$$

for any x .

and [definition 4.4](#), that

Definition. Suppose X is a random variable. The *probability distribution function* of X is given by

$$\begin{aligned} F_X(x) : \mathbb{R} &\rightarrow [0, 1] \\ x &\mapsto \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq x\}) = \mathbb{P}(X \leq x) \end{aligned}$$

When dealing with discrete random variables, it was possible to only consider $\mathbb{P}(X = x)$ for some x . But now, we will have to work in more generality.

Let us observe some important properties.

- ① if $x \leq y$, then $F(x) \leq F(y)$;
- ② for any $a < b$,

$$\mathbb{P}(a < X \leq b) = F(b) - F(a)$$

Proof.

$$\begin{aligned} \mathbb{P}(a < X \leq b) &= \mathbb{P}(X \leq b, X > a) \\ &= \mathbb{P}(X \leq b) - \mathbb{P}(X \leq a) \\ &= F(b) - F(a) \end{aligned}$$

■

Lecture 15 – 25/02/26

Proposition 8.1

A probability distribution function F is right continuous, and always has left limits satisfying

$$\lim_{y \rightarrow x^-} F(y) \leq F(x)$$

Proof. Let (x_n) be a decreasing sequence satisfying $x_n \rightarrow x$ as $n \rightarrow \infty$. We want to show $F(x_n) \rightarrow F(x)$ as $n \rightarrow \infty$.

Let $A_n = \{x < X \leq x_n\}$, so $\mathbb{P}(A_n) = F(x_n) - F(x)$ by the previous. Furthermore, $A_{n+1} \subseteq A_n$ – the sequence is decreasing – so $\mathbb{P}(A_n) \rightarrow \mathbb{P}(\bigcap_n A_n)$ by continuity of probability measures. But $\bigcap_n A_n = \emptyset$, so $\mathbb{P}(A_n) \rightarrow 0$. Hence $\mathbb{P}(A_n) = F(x_n) - F(x) \rightarrow 0$ as $n \rightarrow \infty$.

Since F is increasing (and bounded), left limits always exist. That $\lim_{y \rightarrow x^-} F(y) \leq F(x)$ follows again from the fact F is increasing. ■

$$\textcircled{3} \quad F(x-) = \lim_{y \rightarrow x^-} F(y) = \mathbb{P}(X < x).$$

Proof. Clearly $F(x-) = \lim_{n \rightarrow \infty} F(x - \frac{1}{n})$. Define $B_n = \{X \leq x - \frac{1}{n}\}$, so $B_n \subseteq B_{n+1}$. Then

$$\mathbb{P}(B_n) \rightarrow \mathbb{P}\left(\bigcup_n B_n\right) = \mathbb{P}(X < x)$$

since $\bigcup_n B_n = \{X < x\}$. ■

$$\textcircled{4} \quad \lim_{x \rightarrow \infty} F(x) = 1 \text{ and } \lim_{x \rightarrow -\infty} F(x) = 0.$$

Definition 8.2 (Continuous random variables). A random variable X is called *continuous* if its probability distribution F is a continuous function.

Some important properties equivalent to continuity include (fixing $x \in \mathbb{R}$):

- $F(x-) = F(x)$;
- $\mathbb{P}(X < x) = \mathbb{P}(X \leq x)$;
- $\mathbb{P}(X = x) = 0$.

From now on we will consider only continuous random variables satisfying additionally that F is differentiable.

Definition 8.3 (Probability density functions). The *probability density function* of a random variable X is $f(x) = F'(x)$, where F is its probability distribution.

Let us now consider some important properties of pdf's:

- $f \geq 0$;
- $\int_{-\infty}^{\infty} f(x) dx = 1$;

Proposition 8.4

For every $A \subseteq \mathbb{R}$,

$$\mathbb{P}(X \in A) = \int_A f(x) dx$$

Remark. This is a generalisation of the semi-obvious fact that

$$F(x) = \int_{-\infty}^x f(y) dy$$

What is the intuitive meaning of the probability density function? For a small Δx ,

$$\mathbb{P}(x < X \leq x + \Delta x) = \int_x^{x+\Delta x} f(y) dy \approx f(x)\Delta x$$

so we can think of $f(x)$ as being proportional to the probability that X is close to x .

Let us consider some important examples of continuous probability distributions.

- ① *Uniform distribution* on $[a, b]$, $a < b$. We have

$$f(x) = \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & \text{o/w} \end{cases}$$

$f \geq 0$ and $\int_{-\infty}^{\infty} f(x) dx = 1$, so f is indeed a pdf. Let X be a random variable with density f , denoted $X \sim [a, b]$. Then, for $x \in [a, b]$,

$$\mathbb{P}(X \leq x) = \int_{-\infty}^x f(y) dy = \frac{x-a}{b-a}$$

Also, if $x > b$, $\mathbb{P}(X \leq x) = 1$, and if $x < a$, $\mathbb{P}(X \leq x) = 0$. A special case is the *standard uniform distribution* $X \sim U(0, 1)$

- ② *Exponential distribution*. Let $\lambda > 0$, $f(x) = \lambda e^{-\lambda x}$, $x > 0$. Then

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^{\infty} \lambda e^{-\lambda x} dx = 1$$

so f is a density. letting X have density f , we write $X \sim \text{Exp}(\lambda)$. We have

$$\begin{aligned} \mathbb{P}(X \leq x) &= \int_0^x f(y) dy = 1 - e^{-\lambda x} \\ \mathbb{P}(X \geq x) &= e^{-\lambda x} \end{aligned}$$

Let $T \sim \text{Exp}(\lambda)$, $n \in \mathbb{N}$, $T_n = \lfloor nT \rfloor$. What is the distribution of T_n ?

Let $k \in \mathbb{N}$, and let us try to evaluate $\mathbb{P}(T_n \geq k) = \mathbb{P}(nT \geq k) = \mathbb{P}(T \geq \frac{k}{n}) = e^{-\lambda \frac{k}{n}} = \left(e^{-\frac{\lambda}{n}}\right)^k$. So T_n has the geometric distribution with success parameter $p = 1 - e^{-\frac{\lambda}{n}} \approx \frac{\lambda}{n}$ for n large. In particular,

$$\frac{T_n}{n} \rightarrow T$$

as $n \rightarrow \infty$ – we can obtain the exponential random variable as a scaled limit of geometrics.

The exponential distribution has a cool property! Let $T \sim \text{Exp}(\lambda)$, and let $t, s \in \mathbb{R}^+$. Then

$$\begin{aligned} \mathbb{P}(T \geq s+t | T \geq s) &= \frac{\mathbb{P}(T \geq s+t | T \geq s)}{\mathbb{P}(T \geq s)} \\ &= \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} \end{aligned}$$

so $\mathbb{P}(T \geq s+t | T \geq s) = \mathbb{P}(T \geq t)$. This is called the *memoryless* property (the geometric distribution also has it).

Proposition 8.5

Let T be a positive random variable which is not identically zero or infinity. Then T has the exponential distribution if and only if T has the memoryless property.

Proof. Suppose T has the memoryless property. Set $g(t) = \mathbb{P}(T \geq t)$, so

$$g(t+s) = g(t)g(s)$$

Therefore, $g(mt) = g(t)^m$ for any $t > 0$, $m \in \mathbb{N}$. Putting $t = 1$, we have $g(m) = g(1)^m$, and we will define $\lambda = -\log \mathbb{P}(T \geq 1)$ such that $g(1) = \mathbb{P}(T \geq 1) = e^{-\lambda}$. Hence $g(m) = e^{-\lambda m}$ for all $m \in \mathbb{N}$, and, for $m, n \in \mathbb{N}$, $(g(\frac{m}{n}))^n = g(m)$, so $g(q) = e^{-\lambda q}$ for rationals q . How do we extend this to the reals?

Let $t > 0$, and let $r, s \in \mathbb{Q}^+$ such that $s < t < r$ and $|s - r| \leq \varepsilon$, which can be done by density of the rationals. Then

$$\underbrace{g(r)}_{e^{-\lambda r}} \leq g(t) \leq \underbrace{g(s)}_{e^{-\lambda s}}$$

Taking $\varepsilon \rightarrow 0$, we get $g(t) = e^{-\lambda t}$ as required. ■

Lecture 16 – 27/02/26

Definition 8.6. The *expectation* of a non-negative continuous random variable X , $X \geq 0$, with density f , is given by

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x) dx$$

and

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$$

Definition 8.7. The expectation of a general continuous random variable X with density f is defined as follows. We put

$$\begin{aligned} X_+ &= \max(X, 0) \\ X_- &= \max(-X, 0) \end{aligned}$$

so $X = X_+ - X_-$. If $\mathbb{E}[X_+]$ or $\mathbb{E}[X_-] < \infty$, we define

$$\mathbb{E}[X] = \mathbb{E}[X_+] - \mathbb{E}[X_-] = \int_{-\infty}^{\infty} x f(x) dx$$

Lemma 8.8

If X is a continuous random variable with $X \geq 0$, then

$$\mathbb{E}[X] = \int_0^{\infty} \mathbb{P}(X \geq x) dx$$

Proof. Observe

$$\begin{aligned}
 \mathbb{E}[X] &= \int_0^\infty x f(x) dx \\
 &= \int_0^\infty \left(\int_0^x 1 dy \right) f(x) dx \\
 &= \int_0^\infty \left(\int_y^\infty f(x) dx \right) dy \\
 &= \int_0^\infty \mathbb{P}(X \geq y) dy
 \end{aligned}$$

■

Remark. Recall in the discrete case we instead took expectations of the equality

$$X = \sum_{n=1}^{\infty} 1(X \geq n)$$

Can we do something similar for continuous random variables? Indeed we have

$$\begin{aligned}
 X(\omega) &= \int_0^\infty 1(X(\omega) \geq x) dx \\
 \implies \mathbb{E}[X] &= \mathbb{E} \left[\int_0^\infty 1(X \geq x) dx \right] \\
 &= \int_0^\infty \mathbb{P}(X \geq x) dx
 \end{aligned}$$

However, the interchange of expected value and integral is not justified.

Example

Let us consider some examples.

- Take $X \sim U(a, b)$. Then

$$\mathbb{E}[X] = \int_a^b x f(x) dx = \frac{a+b}{2}$$

- Take $X \sim \text{Exp}(\lambda)$. Then

$$\mathbb{E}[X] = \int_0^\infty \lambda x e^{-\lambda x} dx = \frac{1}{\lambda}$$

or, perhaps more easily,

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X \geq x) dx = \int_0^\infty e^{-\lambda x} dx$$

§8.1 The normal distribution

Definition 8.9 (Normal distribution). Let $\mu \in \mathbb{R}$, $\sigma > 0$ be two parameters. Then

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

is the density of the *normal distribution*.

Proof. We must check f is a density. Observe

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) du \\ &= 2 \int_0^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) du \end{aligned}$$

Let this integral be I . We find

$$\begin{aligned} I^2 &= \frac{2}{\pi} \int_0^{\infty} \int_0^{\infty} \exp\left(-\frac{u^2}{2}\right) \exp\left(-\frac{v^2}{2}\right) du dv \\ &= \frac{2}{\pi} \int_0^{\infty} \int_0^{\pi/2} r e^{-\frac{r^2}{2}} d\theta dr \\ &= \int_0^{\infty} r e^{-\frac{r^2}{2}} dr = 1 \end{aligned}$$

Since $f(x) > 0$, we are done. ■

Let X be a random variable with density f . We have

$$\begin{aligned} \mathbb{E}[X] &= \int_{-\infty}^{\infty} x f(x) dx \\ &= \underbrace{\int_{-\infty}^{\infty} (x-\mu) \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx}_{\text{odd around } \mu} + \underbrace{\int_{-\infty}^{\infty} \frac{\mu}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx}_{\mu} \\ &= \mu \end{aligned}$$

Furthermore,

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx \\ &= \sigma^2 \underbrace{\int_{-\infty}^{\infty} \frac{u^2}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) du}_{1 \text{ after integration by parts}} \\ &= \sigma^2 \end{aligned}$$

after $u = \frac{x-\mu}{\sigma}$. So $\text{Var}(X) = \sigma^2$.

When X has the normal distribution, we write $X \sim \mathcal{N}(\mu, \sigma^2)$ and we call it a normal random variable. If $X \sim \mathcal{N}(1, 0)$, X is a *standard normal variable*.

Proposition 8.10

Let X have density f . Let g be a strictly monotone function with g^{-1} differentiable. Then $g(X)$ has density given by

$$f(g^{-1}(x)) \left| \frac{d}{dx} g^{-1}(x) \right|$$

Proof. Suppose g is increasing. Then

$$\mathbb{P}(g(X) \leq x) = \mathbb{P}(X \leq g^{-1}(x)) = F(g^{-1}(x))$$

Differentiating,

$$\frac{d}{dx} \mathbb{P}(g(X) \leq x) = \frac{d}{dx} (F(g^{-1}(x))) = f(g^{-1}(x)) \underbrace{\frac{d}{dx} g^{-1}(x)}_{>0}$$

If g is decreasing, we put $\mathbb{P}(g(X) \leq x)$ and proceed analogously. ■

§8.1.1 Linear transformations of a normal random variable

Let $X \sim \mathcal{N}(\mu, \sigma^2)$. Let $a \neq 0$, and $g(x) = ax + b$. What is the density of $Y = g(X)$? Clearly

$$g^{-1}(x) = \frac{x - b}{a} \implies \frac{d}{dx} g^{-1}(x) = \frac{1}{a}$$

Hence

$$\begin{aligned} f_Y(y) &= f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| \\ &= \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{\left(\frac{y-b}{a} - \mu\right)^2}{2\sigma^2}\right) \cdot \frac{1}{|a|} \\ &= \frac{1}{|a|\sigma\sqrt{2\pi}} \exp\left(-\frac{(y - (a\mu + b))^2}{2a^2\sigma^2}\right) \end{aligned}$$

so $Y \sim \mathcal{N}(a\mu + b, (a\sigma)^2)$. Hence, letting $X \sim \mathcal{N}(\mu, \sigma^2)$, we have

$$\frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$$

This is a very helpful result! Say, for example, I have $X \sim \mathcal{N}(\mu, \sigma^2)$, and I want to find $\mathbb{P}(-2\sigma < X - \mu < 2\sigma)$. This is just $\mathbb{P}(-2 < \mathcal{N}(0, 1) < 2)$ for the standard normal variable $\mathcal{N}(0, 1)$.

Definition 8.11 (Standard normal probability distribution function). Let

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) du = \mathbb{P}(\mathcal{N}(0, 1) \leq x)$$

Now

$$\phi(x) = \Phi'(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

so $\phi(x) - \phi(-x) = 0$, and hence $(\Phi(x) + \Phi(-x))' = 0$, so $\Phi(x) + \Phi(-x)$ takes a constant value of 1.

Definition 8.12 (Median). Let X be a continuous random variable. The *median* of X , denoted by m , is the number satisfying

$$\mathbb{P}(X \geq m) = \mathbb{P}(X \leq m) = \frac{1}{2}$$

or

$$\int_{-\infty}^m f(x) dx = \int_m^{\infty} f(x) dx = \frac{1}{2}$$

Example

Suppose $X \sim \mathcal{N}(\mu, \sigma^2)$. We claim the median is μ . Observe

$$\mathbb{P}(X \leq \mu) = \mathbb{P}\left(\frac{X - \mu}{\sigma} \leq 0\right) = \mathbb{P}(\mathcal{N}(0, 1) \leq 0) = \frac{1}{2}$$

§8.2 Multivariate density functions

In the discrete case, we defined the joint distribution of $X_1, \dots, X_n$ in terms of

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n)$$

However, in the continuous case, such a probability is of course zero, so we cannot talk about this. Instead, we say

Definition 8.13 (Joint probability distribution function). The *probability distribution function* F for a vector $X = (X_1, \dots, X_n)$, where X_i are random variables, is such that

$$F(x_1, \dots, x_n) = \mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n)$$

Definition 8.14 (Joint probability density function). The *probability density function* f for a vector $X = (X_1, \dots, X_n)$, where X_i are random variables, is such that

$$f(x_1, \dots, x_n) = \frac{\partial^n F}{\partial x_1 \dots \partial x_n}(x_1, \dots, x_n)$$

or, equivalently,

$$\mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n) = \int_{-\infty}^{x_1} \dots \int_{-\infty}^{x_n} f(y_1, \dots, y_n) dy_n \dots dy_1$$

Proposition 8.15

Let $B \subset \mathbb{R}^n$. Then

$$\mathbb{P}((X_1, \dots, X_n) \in B) = \int_B f(y_1, \dots, y_n) dy_1 \dots dy_n$$

Definition 8.16 (Independence). Let $X_1, \dots, X_n$ be random variables. We say they are independent if, for any $x_1, \dots, x_n$,

$$\mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n) = \mathbb{P}(X_1 \leq x_1) \dots \mathbb{P}(X_n \leq x_n)$$

Proposition 8.17

Suppose $X = (X_1, \dots, X_n)$ has density f . Then

- Suppose $X_1, \dots, X_n$ are independent and have densities given by $f_1, \dots, f_n$. Then $f(x_1, \dots, x_n) = f_1(x_1) \dots f_n(x_n)$.
- Suppose $f(x_1, \dots, x_n) = f_1(x_1) \dots f_n(x_n)$ for some non-negative functions $f_1, \dots, f_n$. Then $X_1, \dots, X_n$ are independent with densities proportional to each of $f_1, \dots, f_n$.

Proof. • We have

$$\begin{aligned}\mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n) &= \mathbb{P}(X_1 \leq x_1) \dots \mathbb{P}(X_n \leq x_n) \\ &= \left(\int_{-\infty}^{x_1} f_1(y_1) dy_1 \right) \dots \left(\int_{-\infty}^{x_n} f_n(y_n) dy_n \right) \\ &= \int_{-\infty}^{x_1} \dots \int_{-\infty}^{x_n} f_1(y_1) \dots f_n(y_n) dy_n \dots dy_1\end{aligned}$$

so $f(x_1, \dots, x_n) = f_1(x_1) \dots f_n(x_n)$ as required.

• We have

$$\begin{aligned}\mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n) &= \int_{-\infty}^{x_1} \dots \int_{-\infty}^{x_n} f(y_1, \dots, y_n) dy_n \dots dy_1 \\ &= \left(\int_{-\infty}^{x_1} f_1(y_1) dy_1 \right) \dots \left(\int_{-\infty}^{x_n} f_n(y_n) dy_n \right)\end{aligned}$$

Observe $\int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f_1(y_1) \dots f_n(y_n) dx_1 \dots dx_n = 1$. Hence

$$\mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n) = \frac{\int_{-\infty}^{x_1} f_1(y_1) dy_1}{\int_{-\infty}^{\infty} f_1(y_1) dy_1} \dots \frac{\int_{-\infty}^{x_n} f_n(y_n) dy_n}{\int_{-\infty}^{\infty} f_n(y_n) dy_n}$$

So $X_1, \dots, X_n$ are independent, with X_i having density

$$\frac{f(x_i)}{\int_{-\infty}^{\infty} f(y_i) dy_i}$$

■

§8.2.1 Marginal density

Definition 8.18 (Marginal density). Let $X = (X_1, \dots, X_n)$ be a vector of random variables. The *marginal density* of X_i , $f_{X_i}(x_i)$, is the density of X_i , where all other variables' values are ignored.

Proposition 8.19

Let $X = (X_1, \dots, X_n)$ be a vector of random variables. The marginal density is given by

$$f_{X_i}(x_i) = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f(x_1, \dots, x_i, \dots, x_n) dx_n \dots dx_{i+1} dx_{i-1} \dots dx_1$$

Proof. Without loss of generality take $i = 1$. Then

$$\begin{aligned}\mathbb{P}(X_1 \leq x) &= \mathbb{P}(X_1 \leq x, X_2 \in \mathbb{R}, \dots, X_n \in \mathbb{R}) \\ &= \int_{-\infty}^x \left(\int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f(x_1, \dots, x_n) dx_n \dots dx_2 \right) dx_1\end{aligned}$$

so the expression inside the integral must be the density of X_1 . ■

§8.2.2 Sum of independent random variables

Proposition 8.20

Suppose X and Y are independent random variables with density f_X and f_Y respectively. The density of $X + Y$ is given by

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx$$

Proof. Observe

$$\begin{aligned} \mathbb{P}(X + Y \leq z) &= \mathbb{P}((X, Y) \in \{(x, y) \in \mathbb{R}^2 : x + y \leq z\}) \\ &= \int_{x+y \leq z} f_{X,Y}(x, y) dx dy \\ &= \int_{x+y \leq z} f_X(x) f_Y(y) dx dy \\ &= \int_{-\infty}^{\infty} f_X(x) \int_{-\infty}^{z-x} f_Y(y) dy \\ &= \int_{-\infty}^{\infty} f_X(x) \int_{-\infty}^z f_Y(y - x) dy dx \\ &= \int_{-\infty}^z \left(\int_{-\infty}^{\infty} f_X(x) f_Y(y - x) dx \right) dy \end{aligned}$$

so the density of $X + Y$ is as required. ■

§8.2.3 Conditional density

Definition 8.21 (Conditional density). Let X and Y be random variables with joint density $f_{X,Y}$ and marginal densities f_X and f_Y . The *conditional density* of X given $Y = y$ is

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

for $f_Y(y) > 0$.

Proposition 8.22 (Law of total probability for continuous random variables)

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy = \int_{-\infty}^{\infty} f_{X|Y}(x|y) f_Y(y) dy$$

Definition 8.23 (Conditional expectation for continuous random variables). We define $\mathbb{E}[X|Y]$, the *conditional expectation* of X given y , to be $g(Y)$, where

$$g(y) = \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx$$

Theorem 8.24

Let X be a random variable with values in $D \subseteq \mathbb{R}^n$ and density f . Let g be a bijection from D to $g(D)$ with continuous derivatives on D and $|J| \neq 0$ for any $x \in D$, where $|J|$ is the Jacobian of g^{-1} . Then, the random variable $Y = g(X)$ has density

$$f_Y(y) = f_X(g^{-1}(y)) |J|$$

Example

Suppose X, Y are independent and $X, Y \sim \mathcal{N}(0, 1)$. What are the distributions of the magnitude and argument of $X + iY$? By the above,

$$\begin{aligned} f_{R,\theta}(r, \theta) &= f_{X,Y}(r \cos \theta, r \sin \theta) |J| = f_X(r \cos \theta) f_Y(r \sin \theta) r \\ &= \frac{1}{\sqrt{2\pi}} e^{-\frac{r^2 \cos^2 \theta}{2}} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{r^2 \sin^2 \theta}{2}} r \\ &= \frac{1}{2\pi} r e^{-\frac{r^2}{2}} \end{aligned}$$

Hence, $\theta \sim U(0, 2\pi)$, R has density $r e^{-\frac{r^2}{2}}$, and they are independent.

§8.3 Order statistics of a random sample

Let $X_1, \dots, X_n$ be i.i.d. random variables with probability distribution function F and density f . For a given sample, let $Y_1 \leq \dots \leq Y_n$ be the X_i ordered from smallest to largest – called the order statistics – and let us try to find the density of $(Y_1, \dots, Y_n)$.

Firstly, observe

$$\mathbb{P}(Y_1 \leq x) = 1 - \mathbb{P}(Y_1 > x) = 1 - \mathbb{P}(X_1 > x)^n = 1 - (1 - F(x))^n$$

by independence. Differentiating, we obtain

$$f_{Y_1}(x) = \frac{d}{dx} \mathbb{P}(Y_1 \leq x) = n f(x) (1 - F(x))^{n-1}$$

Similarly,

$$\begin{aligned} \mathbb{P}(Y_n \leq x) &= F(x)^n \\ \implies f_{Y_n}(x) &= n f(x) F(x)^{n-1} \end{aligned}$$

Can we generalise these results? Yes!

Proposition 8.25

Let $X_1, \dots, X_n$ be i.i.d. random variables with probability distribution function F and density f , and let $Y_1, \dots, Y_n$ be their order statistics. Then

$$f_{(Y_1, \dots, Y_n)}(x_1, \dots, x_n) = \begin{cases} n! f(x_1) \dots f(x_n) & x_1 < \dots < x_n \\ 0 & \text{o/w} \end{cases}$$

Proof. Considering all permutations of the X_i which give the same ordered Y_i 's, we have

$$\begin{aligned}
 & \mathbb{P}(Y_1 \leq x_1, \dots, Y_n \leq x_n) \\
 &= n! \cdot \mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n, X_1 \leq \dots \leq X_n) \\
 &= n! \int_{-\infty}^{x_1} \int_{u_1}^{x_2} \dots \int_{u_{n-1}}^{x_n} f(u_n) f(u_{n-1}) \dots f(u_1) du_n \dots du_1 \\
 &\implies f_{(Y_1, \dots, Y_n)}(x_1, \dots, x_n) = \begin{cases} n! f(x_1) \dots f(x_n) & x_1 < \dots < x_n \\ 0 & \text{o/w} \end{cases}
 \end{aligned}$$

after differentiation and the application of FTC. Notably, the Y_i are not independent, since the joint distribution only factorises in the case $x_1 < \dots < x_n$. ■

Example

Let X_i be independent with $X_i \sim \text{Exp}(\lambda_i)$, and let $X = \min(X_1, \dots, X_n)$. Then

$$\begin{aligned}
 \mathbb{P}(X \leq x) &= 1 - \left(1 - \left(1 - e^{-\lambda_1 x}\right)\right) \dots \left(1 - \left(1 - e^{-\lambda_n x}\right)\right) \\
 &= 1 - e^{-(\lambda_1 + \dots + \lambda_n)x}
 \end{aligned}$$

so $X \sim \text{Exp}(\sum_{i=1}^n \lambda_i)$. Now let $Y_1, \dots, Y_n$ be the order statistics, and set $Z_1 = Y_1$, $Z_2 = Y_2 - Y_1$, $Z_i = Y_i - Y_{i-1}$ for $i \leq n$ – so the Z_i 's represent the spacing between the samples. What is the joint density of $(Z_1, \dots, Z_n)$?

Observe

$$\underbrace{\begin{pmatrix} Z_1 \\ \vdots \\ Z_n \end{pmatrix}}_{\mathbf{Z}} = \underbrace{\begin{pmatrix} 1 & 0 & 0 & \dots & 0 & 0 \\ -1 & 1 & 0 & \dots & 0 & 0 \\ 0 & -1 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 \end{pmatrix}}_A \underbrace{\begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}}_{\mathbf{Y}}$$

and also that

$$y_j = \sum_{i=1}^j z_i$$

Now, by the theorem,

$$\begin{aligned}
 f_{(Z_1, \dots, Z_n)}(z_1, \dots, z_n) &= f_{(Y_1, \dots, Y_n)}(y_1, \dots, y_n) |J| \\
 &= n! f(y_1) \dots f(y_n) \\
 &= n! \lambda e^{-\lambda z_1} \lambda e^{-\lambda(z_1+z_2)} \dots \lambda e^{-\lambda(z_1+\dots+z_n)} \\
 &= \prod_{i=1}^n \left((n-i+1) \lambda e^{-\lambda(n-i+1)z_i} \right)
 \end{aligned}$$

so the Z_i 's are independent, and $Z_i \sim \text{Exp}(\lambda(n-i+1))$.

§9 Moment generating functions

Definition 9.1 (Moment generating functions). Let X be a random variable with density f . The *moment generating function* or *mgf* of X is defined to be

$$m(\theta) = \mathbb{E}[e^{\theta X}] = \int_{-\infty}^{\infty} e^{\theta x} f(x) dx$$

whenever this integral is finite.

Theorem 9.2

The mgf uniquely determines the distribution of a random variable, provided it is defined for an open interval of values of θ . Furthermore,

$$m^{(r)}(\theta) = \mathbb{E}[X^r]$$

Example

The *gamma distribution*, for $n \in \mathbb{N}$ and $\lambda > 0$, has density

$$f(x) = \frac{e^{-\lambda x} \lambda^n x^{n-1}}{(n-1)!}$$

which has the distribution of a sum of n i.i.d. exponential random variables. Let us check that it is a density.

$$\begin{aligned} I_n &= \int_0^{\infty} f(x) dx = \int_0^{\infty} \lambda e^{-\lambda x} \cdot \frac{\lambda^{n-1} x^{n-1}}{(n-1)!} dx \\ &= \int_0^{\infty} \lambda e^{-\lambda x} \cdot \frac{\lambda^{n-2} x^{n-2}}{(n-2)!} dx && \text{(parts)} \\ &= I_{n-1} = \dots = I_1 = 1 \end{aligned}$$

Suppose $X \sim \Gamma(n, \lambda)$. What is the mgf of X ? We have

$$\begin{aligned} m(\theta) &= \int_0^{\infty} e^{\theta x} e^{-\lambda x} \frac{\lambda^n x^{n-1}}{(n-1)!} dx \\ &= \frac{\lambda^n}{(\lambda - \theta)^n} \int_0^{\infty} e^{-(\lambda - \theta)x} \frac{(\lambda - \theta)^n x^{n-1}}{(n-1)!} dx \end{aligned}$$

If $\theta < \lambda$, then

$$m(\theta) = \left(\frac{\lambda}{\lambda - \theta} \right)^n$$

Proposition 9.3

If $X_1, \dots, X_n$ are independent, then $X_1 + \dots + X_n$ has mgf

$$m(\theta) = \mathbb{E}[e^{\theta(X_1 + \dots + X_n)}] = \prod_{i=1}^n \mathbb{E}[e^{\theta X_i}]$$

Example (Sum of gamma random variables)

Suppose $X \sim \Gamma(n, \lambda)$ and $Y \sim \Gamma(m, \lambda)$, and $X \perp Y$. Then

$$\mathbb{E}[e^{\theta(X+Y)}] = \mathbb{E}[e^{\theta X}] \mathbb{E}[e^{\theta Y}] = \left(\frac{\lambda}{\lambda - \theta} \right)^n \left(\frac{\lambda}{\lambda - \theta} \right)^m = \left(\frac{\lambda}{\lambda - \theta} \right)^{n+m}$$

so $X + Y \sim \Gamma(n + m, \lambda)$. It is a consequence of this that, if $X_1, \dots, X_n$ are i.i.d. $\text{Exp}(\lambda)$ random variables, then $X_1 + \dots + X_n \sim \Gamma(n, \lambda)$.

Remark. One can generalise the gamma distribution to non-integer m by replacing $(n - 1)!$ with

$$\Gamma(\alpha) = \int_0^\infty e^{-x} x^{\alpha-1} dx$$

so that $X \sim \Gamma(\alpha, \lambda)$ if

$$f(x) = \frac{e^{-\lambda x} \lambda^\alpha x^{\alpha-1}}{\Gamma(\alpha)}$$

Definition 9.4 (Cauchy distribution). The *Cauchy distribution* has density

$$f(x) = \frac{1}{\pi(1 + x^2)}$$

Hence, the mgf is

$$m(\theta) = \int_{-\infty}^{\infty} \frac{e^{\theta x}}{\pi(1 + x^2)} dx = \infty$$

Suppose $X \sim f$, then observe $X, 2X, \dots$ all have the same mgf, but not the same distribution. Hence mgfs do not uniquely determine distributions unless they are finite.

What is the mgf of a normally distributed random variable? We have

$$\begin{aligned} m(\theta) &= \int_{-\infty}^{\infty} e^{\theta x} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2} + 2\frac{x}{2\sigma^2}(\mu + \theta\sigma^2) - \frac{\mu^2}{2\sigma^2}\right) dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2}(x - (\mu + \theta\sigma^2))^2 + \mu\theta + \frac{\theta^2\sigma^2}{2}\right) dx \\ &= \exp\left(\mu\theta + \frac{\theta^2\sigma^2}{2}\right) \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2}(x - (\mu + \theta\sigma^2))^2\right) dx \end{aligned}$$

Then, the integrand is the density of $\mathcal{N}(\mu + \theta\sigma^2, \sigma^2)$, so we obtain

$$m(\theta) = \exp\left(\mu\theta + \frac{\theta^2\sigma^2}{2}\right)$$

This gives an easy way to see that

Proposition 9.5

If $X \sim \mathcal{N}(\mu, \sigma^2)$, $Y \sim \mathcal{N}(\nu, \tau^2)$ and $X \perp Y$, then $X + Y \sim \mathcal{N}(\mu + \nu, \sigma^2 + \tau^2)$.

§9.1 Multivariate moment generating functions

Definition 9.6 (Multivariate moment generating functions). Let $X = (X_1, \dots, X_n)$ be a random variable. The mgf of X is defined to be

$$m(\theta) = \mathbb{E}[e^{\theta^T X}]$$

Theorem 9.7

If the mgf is finite for an open set of values of θ , then it uniquely determines the distribution – furthermore,

$$\left. \frac{\partial^r m}{\partial \theta_i^r} \right|_{\theta=0} = \mathbb{E}[X_i^r], \quad \left. \frac{\partial^{r+s} m}{\partial \theta_i^r \partial \theta_j^s} \right|_{\theta=0} = \mathbb{E}[X_i^r X_j^s]$$

Proposition 9.8

Let $(X_1, \dots, X_n) = X$ be a random variable in $\mathbb{R}^n$. Then

$$m(\theta) = \prod_{i=1}^n \mathbb{E}[e^{\theta_i X_i}]$$

if and only if $X_1, \dots, X_n$ are independent.

Proof. If $X_1, \dots, X_n$ are independent, then $m(\theta)$ factorises. Conversely, if $m(\theta)$ factorises, then the X_i must be independent by the uniqueness of mgfs. ■

§9.2 Multivariate Gaussian random variables

Definition 9.9 (Multivariate Gaussian random variables). Let $X = (X_1, \dots, X_n)$ be a random variable. We say X is *Gaussian* if, for any $u \in \mathbb{R}^n$, $u^T X$ is a Gaussian random variable in $\mathbb{R}$.

Proposition 9.10

Let $X \in \mathbb{R}^n$ be a Gaussian vector. Let A be an m by n matrix and let $b \in \mathbb{R}^m$. Then $AX + b$ is also Gaussian.

Proof. Let $u \in \mathbb{R}^m$ – we want to show $u^\top (AX + b)$ is Gaussian in $\mathbb{R}$. Observe

$$\begin{aligned} u^\top (AX + b) &= u^\top AX + u^\top b \\ &= v^\top X + u^\top b \end{aligned}$$

where $v^\top X$ is Gaussian (since X is) and $u^\top b$ is constant. The result follows. ■

Definition 9.11 (Multivariate mean and variance). A random vector variable X has mean

$$\mu = \mathbb{E}[X] = \begin{pmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_n] \end{pmatrix}$$

and variance

$$V = \text{Var}(X) = \mathbb{E}[(X - \mu) \cdot (X - \mu)^\top]$$

which is an n -by- n symmetric matrix, where expectation is taken pointwise.

Remark. Observe that $\text{Var}(X)_{i,j} = \text{Cov}(X_i, X_j)$.

Lecture $n + 1$ – 09/03/26

What is the expectation of $u^\top X$? It should be

$$\mathbb{E} \left[\sum_{i=1}^n u_i X_i \right] = \sum_{i=1}^n u_i \mathbb{E}[X_i] = u^\top \mu$$

Furthermore,

$$\text{Var}(u^\top X) = \text{Var} \left(\sum_{i=1}^n u_i X_i \right) = \sum_{i,j=1}^n u_i \text{Cov}(X_i, X_j) u_j = u^\top V u$$

Proposition 9.12

V is a non-negative definite matrix – for any $u \in \mathbb{R}^n$,

$$u^\top V u \geq 0$$

Proof. $u^\top V u = \text{Var}(u^\top X) \geq 0$. ■

Let us also find the mgf. We have, for any vector λ , $\lambda^\top X \sim \mathcal{N}(\lambda^\top \mu, \lambda^\top V \lambda)$. It follows immediately from the expression for the mgf of a Gaussian random variable in $\mathbb{R}$ that

$$m(\lambda) = \exp \left(\lambda^\top \mu + \frac{\lambda^\top V \lambda}{2} \right)$$

Recall that the mgf uniquely defines the distribution in any case – hence, for a Gaussian random variable in particular, we conclude μ and V are sufficient.

Lemma 9.13 (Construction of a Gaussian random variable)

Let $Z_1, \dots, Z_n$ be i.i.d $\mathcal{N}(0, 1)$ random variables. Then $Z = (Z_1, \dots, Z_n)$ is a Gaussian vector.

Proof. Let $u \in \mathbb{R}^n$, and we need to show $u^\top Z$ is normal in $\mathbb{R}$. We have that the mgf of $u^\top Z$ is

$$\begin{aligned} m(\lambda) &= \mathbb{E} \left[e^{\lambda \sum_{i=1}^n u_i Z_i} \right] \\ &= \prod_{i=1}^n \mathbb{E} \left[e^{\lambda u_i Z_i} \right] \\ &= \prod_{i=1}^n \exp \left(\frac{\lambda^2 u_i^2}{2} \right) \\ &= \exp \left(\frac{\lambda^2 |u|^2}{2} \right) \end{aligned}$$

Hence, $u^\top Z \sim \mathcal{N}(0, |u|^2)$. ■

Remark. For this vector, we have in particular $\mu = \mathbf{0}$, $V = I_n$ – we write $Z \sim \mathcal{N}(0, I_n)$.

Let $\mu \in \mathbb{R}^n$ and let V be a non-negative definite matrix. We want to construct a Gaussian vector with mean μ and variance V , using the standard Gaussian Z . Recall that, for $n = 1$, we can do this by taking $X = \mu + \sigma Z$. However, for this, we need to somehow take the square root of the variance – how will this generalise?

Definition 9.14 (Matrix square root). Let V be a non-negative definite matrix such that

$$V = U^\top D U$$

where $U^\top = U^{-1}$ and D is a diagonal matrix, which has diagonal entries all ≥ 0 . Then

$$\sqrt{V} = U^\top \sqrt{D} U$$

where the square root of D is produced by taking the square root of each of its diagonal entries.

Remark. Reassuringly, this definition satisfies $(\sqrt{V})^2 = V$.

Lemma 9.15 (Construction of a general Gaussian random variable)

Let $\mu \in \mathbb{R}^n$ and V be a non-negative definite matrix. Let $Z_1, \dots, Z_n$ be i.i.d $\mathcal{N}(0, 1)$ random variables and $Z = (Z_1, \dots, Z_n)$. Let $X = \mu + \sigma Z$, where σ is the square root of V . Then X is Gaussian with mean μ and variance $V = \sigma^2$, denoted $X \sim \mathcal{N}(\mu, V)$.

Proof. X is Gaussian by [proposition 9.10](#). Clearly $\mathbb{E}[X] = \mu$, and

$$\text{Var}(X) = \mathbb{E}[(X - \mu)(X - \mu)^\top]$$

$$\begin{aligned}
&= \mathbb{E}[(\sigma Z)(Z^\top \sigma^\top)] \\
&= \sigma \mathbb{E}[ZZ^\top] \sigma^\top \\
&= \sigma I_n \sigma^\top \\
&= V
\end{aligned}$$

■

Proposition 9.16

Let $X \sim \mathcal{N}(\mu, V)$, where V is positive definite. Then, the density of X is given by

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^n \det V}} \exp\left(-\frac{(x - \mu)^\top V^{-1}(x - \mu)}{2}\right)$$

Proof. We can write $X = \mu + \sigma Z$, where $Z \sim \mathcal{N}(0, I_n)$, and $Z = \sigma^{-1}(x - \mu)$. Hence, by [theorem 8.24](#), we have

$$\begin{aligned}
f_X(x) &= f_Z(z) |J| \\
&= \frac{1}{(2\pi)^{n/2}} \exp\left(-\frac{z^\top z}{2}\right) \frac{1}{\det \sigma} \\
&= \frac{1}{(2\pi)^{n/2} \det \sigma} \exp\left(-\frac{(x - \mu)^\top (\sigma^{-1})^\top \sigma^{-1} (x - \mu)}{2}\right) \\
&= \frac{1}{\sqrt{(2\pi)^n \det V}} \exp\left(-\frac{(x - \mu)^\top V^{-1} (x - \mu)}{2}\right)
\end{aligned}$$

■

We now seek to generalise this result to non-negative definite variance. If V has some eigenvalue equal to zero, there exists an orthogonal change of basis such that

$$V = \begin{pmatrix} U & 0 \\ 0 & 0 \end{pmatrix}$$

where U is an m by m positive definite matrix with all other entries filled in by zero. Then, we can write $\mu = (\lambda, \nu)$ where $\lambda \in \mathbb{R}^m$, $\nu \in \mathbb{R}^{n-m}$, so that $X = (y, \nu)$ for a constant vector ν and $Y \sim \mathcal{N}(\lambda, U)$ with density

$$f_Y(y) = \frac{1}{\sqrt{(2\pi)^m \det U}} \exp\left(-\frac{(y - \lambda)^\top U^{-1} (y - \lambda)}{2}\right)$$

Proposition 9.17

Let $X = (X_1, \dots, X_n)$ be a Gaussian vector, $\mu = \mathbb{E}[X]$, $V = \text{Var}(X)$ positive definite. Then $X_1, \dots, X_n$ are independent if and only if V is diagonal.

The forward implication is true for any random variable. We give two proofs of the converse.

Proof 1. Let

$$V = \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix}$$

Let $\mu = (\mu_1, \dots, \mu_n)$, so that

$$\begin{aligned} f_X(x) &= \frac{1}{\sqrt{(2\pi)^n \det V}} \exp\left(-\frac{(x - \mu)^\top V^{-1}(x - \mu)}{2}\right) \\ &= \frac{1}{\sqrt{(2\pi)^n \det V}} \exp\left(-\sum_{i=1}^n \frac{(x_i - \mu_i)^2}{2\lambda_i}\right) \\ &= \prod_{i=1}^n \frac{1}{\lambda_i \sqrt{2\pi}} \exp\left(-\frac{(x_i - \mu_i)^2}{2\lambda_i}\right) \end{aligned}$$

so X_i are independent as required, and furthermore each is normal with mean μ_i and variance λ_i . ■

Proof 2. Observe

$$\begin{aligned} m(\theta) &= \mathbb{E}[e^{\theta^\top X}] = \exp\left(\theta^\top \mu + \frac{\theta^\top V \theta}{2}\right) \\ &= \exp\left(\sum_{i=1}^n \theta_i \mu_i + \sum_{i=1}^n \frac{\theta_i^2 \lambda_i}{2}\right) \\ &= \prod_{i=1}^n \exp\left(\theta_i \mu_i + \frac{\theta_i^2 \lambda_i}{2}\right) \end{aligned}$$

so the mgf factorises and hence the X_i are independent. ■

§9.3 Bivariate Gaussians

Definition 9.18 (Correlation). Let $X = (X_1, X_2)$ be a random variable in $\mathbb{R}^2$, with $\mu_i = \mathbb{E}[X_i]$, $\sigma_i^2 = \text{Var}(X_i)$. Then the *correlation* is given by

$$\rho = \text{Corr}(X_1, X_2) = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{Var}(X_1) \text{Var}(X_2)}}$$

Lemma 9.19

For any X , $-1 \leq \rho \leq 1$.

Proof. Cauchy-Schwarz. ■

Lemma 9.20

The variance matrix of X , which is given by

$$V = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

is positive semi-definite.

Proof. Let $u = (u_1, u_2)^\top$, so

$$u^\top V u = (1 \pm \rho)(\sigma_1^2 u_1^2 + \sigma_2^2 u_2^2) \mp \rho(\sigma_1 u_1 - \sigma_2 u_2)^2$$

Depending on whether $\rho \in [-1, 0]$ or $\rho \in [0, 1]$, we take the first plus sign or the first minus sign, respectively. ■

Proposition 9.21 (Conditional expectation for bivariate Gaussians)

Suppose X is a bivariate Gaussian, then

$$\mathbb{E}[X_2 \mid X_1] = \mu_2 + a(X_1 - \mu_1)$$

where $a = \frac{\text{Cov}(X_1, X_2)}{\text{Var}(X_1)}$.

Proof. Let $Y = X_2 - aX_1$. Observe

$$\begin{pmatrix} X_1 \\ Y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -a & 1 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$$

so (X_1, Y) is Gaussian. Observe

$$\begin{aligned} \text{Cov}(X_1, Y) &= \text{Cov}(X_1, X_2 - aX_1) \\ &= \text{Cov}(X_1, X_2) - a \text{Var}(X_1) \end{aligned}$$

Taking $a = \frac{\text{Cov}(X_1, X_2)}{\text{Var}(X_1)}$ gives $\text{Cov}(X_1, Y) = 0$, so X_1 and Y are independent. Then

$$\begin{aligned} \mathbb{E}[X_2 \mid X_1] &= \mathbb{E}[X_2 - aX_1 \mid X_1] + \mathbb{E}[aX_1 \mid X_1] \\ &= \mathbb{E}[X_2 - aX_1] + aX_1 \\ &= \mu_2 - a\mu_1 + aX_1 \end{aligned}$$

■

§10 Convergence results and limit theorems

Definition 10.1 (Convergence in probability). A sequence of random variables (X_n) converges to X in probability, $X_n \xrightarrow{\mathbb{P}} X$ if, for any $\varepsilon > 0$,

$$\mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0$$

as $n \rightarrow \infty$.

Theorem 10.2 (Weak law of large numbers)

Let (X_n) be a sequence of i.i.d. random variables with finite mean. Let $S_n = X_1 + \cdots + X_n$, then

$$\frac{S_n}{n} \xrightarrow{\mathbb{P}} \mu$$

as $n \rightarrow \infty$.

Proof. Assume $\sigma^2 = \text{Var}(X_1) < \infty$. Observe

$$\begin{aligned} \mathbb{P}\left(\left|\frac{S_n}{n} - \mu\right| > \varepsilon\right) &= \mathbb{P}(|S_n - n\mu| > n\varepsilon) \\ &\leq \frac{\text{Var}(S_n)}{n^2\varepsilon^2} \\ &= \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$. ■

Definition 10.3 (Almost sure/probability 1 convergence). Let (X_n) be a sequence of random variables and X a random variable. We say that $X_n \rightarrow X$ *almost surely* or *with probability 1* if

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} X_n = X\right) = 1$$

Perhaps it is most helpful to return to the definition of random variables. We have that there exists $A \in \mathcal{F}$ such that $\mathbb{P}(A) = 1$ and, for ω in that set, the functions X_n converge pointwise to the function X .

Proposition 10.4

If $X_n \rightarrow 0$ a.s. as $n \rightarrow \infty$, then $X_n \xrightarrow{\mathbb{P}} 0$ as $n \rightarrow \infty$.

Proof. Given an $\varepsilon > 0$, we want to show that $\mathbb{P}(|X_n| \leq \varepsilon) \rightarrow 1$ as $n \rightarrow \infty$. Observe

$$\mathbb{P}(|X_n| \leq \varepsilon) \geq \mathbb{P}\left(\underbrace{\bigcap_{m=n}^{\infty} |X_m| \leq \varepsilon}_{A_n}\right)$$

Then the events A_n are increasing, that is $A_n \subseteq A_{n+1}$. It follows by continuity of probability measures that $\mathbb{P}(A_n)$ is increasing and converges to $\mathbb{P}(\bigcup_n A_n)$ as $n \rightarrow \infty$. Hence

$$\mathbb{P}(A_n) \rightarrow \mathbb{P}\left(\bigcup_n \bigcap_{m=n}^{\infty} |X_m| \leq \varepsilon\right)$$

so that

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n| \leq \varepsilon) \geq \mathbb{P}(\exists n \text{ s.t. } m \geq n \implies |X_m| < \varepsilon)$$

$$\geq \mathbb{P}(\forall \varepsilon > 0, \exists n \text{ s.t. } m \geq n \implies |X_m| < \varepsilon)$$

But this is exactly the set of ω such that X_n converges pointwise to X , so it has probability 1 by assumption. Hence

$$\begin{aligned} & \lim_{n \rightarrow \infty} \mathbb{P}(|X_n| \leq \varepsilon) \geq 1 \\ \implies & \lim_{n \rightarrow \infty} \mathbb{P}(|X_n| \leq \varepsilon) = 1 \end{aligned}$$

as required. ■

Remark. In fact convergence in probability does not imply almost sure convergence, as one can show by considering a well-chosen counterexample.

Theorem 10.5 (Strong law of large numbers)

Let (X_n) be a sequence of i.i.d. random variables with finite mean. Let $S_n = X_1 + \dots + X_n$, then

$$\frac{S_n}{n} \rightarrow \mu$$

a.s. as $n \rightarrow \infty$.

Proof (non-examinable). Assume $\sigma^2 = \text{Var}(X_1) < \infty$ and $\mathbb{E}[X_1^4] < \infty$. Let $Y_i = X_i - \mu$, so $\mathbb{E}[Y_i] = 0$, and

$$\mathbb{E}[Y_1^4] = \mathbb{E}[(X_1 - \mu)^4] \leq \mathbb{E}[(2 \max(\mathbb{E}[X_1], \mu))^4] \leq 2^4 (\mathbb{E}[X_1^4] + \mu^4)$$

Set now $S_n = \sum_{i=1}^n Y_i$, and we must show $S_n/n \rightarrow 0$ a.s. as $n \rightarrow \infty$.

We claim that

$$\mathbb{P}\left(\sum_{n=1}^{\infty} \left(\frac{S_n}{n}\right)^4 < \infty\right) = 1$$

It suffices to show that

$$\mathbb{E}\left[\sum \left(\frac{S_n}{n}\right)^4\right] < \infty$$

Observe

$$\begin{aligned} \mathbb{E}\left[\sum \left(\frac{S_n}{n}\right)^4\right] &= \sum \frac{1}{n^4} \mathbb{E}[S_n^4] \\ &= \sum \frac{1}{n^4} \mathbb{E}\left[\sum_{i=1}^n Y_i^4 + \binom{4}{2} \sum_{1 \leq i < j \leq n} Y_i^2 Y_j^2 + R\right] \end{aligned}$$

where R is a sum of terms of the form

$$Y_i^3 Y_j, Y_i^2 Y_j Y_k, Y_i Y_j Y_k Y_l$$

for distinct indices i, j, k, l . For such a term, the expectation factorises by independence, and then, since $\mathbb{E}[X_i] = 0$ and a first power appears in each of them, we obtain 0.

Let us consider the $X_i^2 X_j^2$ terms. We have

$$\mathbb{E}[X_i^2 X_j^2] = \mathbb{E}[Y_i^2]^2$$

Now, by Jensen with $f(x) = x^2$ or just non-negativity of variance, we have $\mathbb{E}[Y_i^2]^2 \leq \mathbb{E}[Y_i^4]$, which is finite. Now Furthermore,

$$\begin{aligned} \sum \frac{1}{n^4} \mathbb{E} \left[\sum_{i=1}^n Y_i^4 + \binom{4}{2} \sum_{1 \leq i < j \leq n} Y_i^2 Y_j^2 \right] &\leq \sum \frac{1}{n^4} \left(n \mathbb{E}[Y_1^4] + 6 \frac{n(n-1)}{2} \mathbb{E}[Y_1^4] \right) \\ &\leq \mathbb{E}[Y_1^4] \sum \frac{1}{n^4} \cdot 3n^2 \\ &= \mathbb{E}[Y_1^4] \cdot \frac{\pi^2}{2} \end{aligned}$$

which is finite, as required. Hence

$$\mathbb{P} \left(\sum_{n=1}^{\infty} \left(\frac{S_n}{n} \right)^4 < \infty \right) = 1$$

so, applying the n^{th} term test, $\left(\frac{S_n}{n} \right)^4 \rightarrow 0$ with probability 1, so taking fourth roots immediately gives the result. ■

§10.1 Central limit theorem

Definition 10.6 (Convergence in distribution). A sequence of random variables (X_n) converges to X in distribution, $X_n \xrightarrow{(d)} X$, if

$$F_{X_n}(x) \rightarrow F_X(x)$$

as $n \rightarrow \infty$ for all $x \in \mathbb{R}$ that are continuity points of F_X .

Lecture 23 – 16/03/26

Proposition 10.7

If $X_n \xrightarrow{\mathbb{P}} X$ as $n \rightarrow \infty$, then $X_n \xrightarrow{(d)} X$ as $n \rightarrow \infty$.

Proof. Let x be a continuity point of F_X , and let $\varepsilon > 0$. Observe that

$$\{X \leq x - \varepsilon\} \subseteq \{X_n \leq x\} \cup \{|X_n - X| > \varepsilon\}$$

and that

$$\{X_n \leq x\} \subseteq \{X \leq x + \varepsilon\} \cup \{|X_n - X| > \varepsilon\}$$

Taking probabilities,

$$\mathbb{P}(X \leq x - \varepsilon) \leq \mathbb{P}(X_n \leq x) + \mathbb{P}(|X_n - X| > \varepsilon)$$

$$\implies F_X(x - \varepsilon) \leq \liminf_{n \rightarrow \infty} \mathbb{P}(X_n \leq x)$$

since the second term goes to zero by assumption. Furthermore,

$$\begin{aligned} \mathbb{P}(X_n \leq x) &\leq \mathbb{P}(X \leq x + \varepsilon) + \mathbb{P}(|X_n - X| > \varepsilon) \\ \implies \limsup_{n \rightarrow \infty} F_{X_n}(x) &\leq F_X(x + \varepsilon) \end{aligned}$$

so that

$$F_X(x - \varepsilon) \leq \liminf F_{X_n}(x) \leq \limsup F_{X_n}(x) \leq F_X(x + \varepsilon)$$

Now, taking $\varepsilon \rightarrow 0$, since x is a continuity point, we obtain

$$\liminf F_{X_n}(x) = \limsup F_{X_n}(x) = \lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$$

■

Proposition 10.8 (Continuity property for mgf's)

Let X_n be random variables with mgfs $m_n(\theta) = \mathbb{E}[e^{\theta X_n}]$ for $\theta \in \mathbb{R}$, and X a random variable with mgf $m(\theta) = \mathbb{E}[e^{\theta X}]$. Suppose $m(\theta) < \infty$ for some $\theta \neq 0$. If $m_n(\theta) \rightarrow m(\theta)$ as $n \rightarrow \infty$ for all $\theta \in \mathbb{R}$, then $X_n \xrightarrow{(d)} X$ as $n \rightarrow \infty$.

We will need this result for the next theorem, although we will not prove it.

Theorem 10.9 (Central limit theorem)

Let X_n be i.i.d. random variables with finite mean μ and finite variance σ^2 . Let $S_n = X_1 + \dots + X_n$, then

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{(d)} Z$$

as $n \rightarrow \infty$, where $Z \sim \mathcal{N}(0, 1)$. In other words,

$$\mathbb{P}\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq x\right) \longrightarrow \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy = \Phi(x)$$

Proof. Let $Y_i = \frac{X_i - \mu}{\sigma}$, so $\mathbb{E}[Y_i] = 0$ and $\text{Var}(Y_i) = 1$. It suffices, then, to show the theorem for a sequence $X_1, \dots, X_n$ i.i.d. with mean 0 and variance 1.

We will unjustifiably assume for our proof that there exists some $\delta > 0$ with $\mathbb{E}[e^{\delta X_1}] + \mathbb{E}[e^{-\delta X_1}]$ is finite.

Let $m(\theta) = \mathbb{E}[e^{\theta X_1}]$. By the continuity property of mgf's, it suffices to show that

$$\mathbb{E}\left[e^{\theta \frac{S_n}{\sqrt{n}}}\right] \longrightarrow \mathbb{E}[e^{\theta Z}] = e^{\frac{\theta^2}{2}}$$

We have

$$\mathbb{E}[e^{\theta \frac{S_n}{\sqrt{n}}}] = \left(m\left(\frac{\theta}{\sqrt{n}}\right)\right)^n$$

$$\begin{aligned}
 &= \left(\mathbb{E} \left[\left(\frac{X_1}{\sqrt{n}} \right)^0 \right] + \mathbb{E} \left[\left(\frac{X_1}{\sqrt{n}} \right)^1 \right] \theta + \mathbb{E} \left[\left(\frac{X_1}{\sqrt{n}} \right)^2 \right] \theta^2 + \dots \right)^n \\
 &= \left(1 + \frac{\theta^2}{2n} + \left| \mathbb{E} \left[\sum_{k \geq 3} \frac{\theta^k}{(\sqrt{n})^k} \cdot \frac{1}{k!} X_1^k \right] \right| \right)^n
 \end{aligned}$$

We now claim that

$$\left| \mathbb{E} \left[\sum_{k \geq 3} \frac{\theta^k}{(\sqrt{n})^k} \cdot \frac{1}{k!} X_1^k \right] \right| = o(|\theta|^2)$$

as $\theta \rightarrow 0$. Let $|\theta| < \delta/2$. Then

$$\begin{aligned}
 \left| \mathbb{E} \left[\sum_{k \geq 3} \frac{\theta^k}{(\sqrt{n})^k} \cdot \frac{1}{k!} X_1^k \right] \right| &\leq \mathbb{E} \left[\sum_{k \geq 3} \frac{|\theta X_1|^k}{k!} \right] \\
 &= \mathbb{E} \left[|\theta X_1|^3 \sum_{k \geq 0} \frac{|\theta X_1|^k}{(k+3)!} \right] \\
 &\leq \mathbb{E} \left[|\theta X_1|^3 \sum_{k \geq 0} \frac{|\theta X_1|^k}{k!} \right] \\
 &= \mathbb{E} \left[|\theta X_1|^3 e^{|\theta X_1|} \right] \\
 &\leq \mathbb{E} \left[|\theta X_1|^3 e^{\frac{\delta |X_1|}{2}} \right]
 \end{aligned}$$

Observe

$$|\theta X_1|^3 = |\theta|^3 \frac{\left| \frac{\delta}{2} X_1 \right|^3}{3!} \cdot \frac{3!}{\left(\frac{\delta}{2} \right)^2} \leq C |\theta|^3 e^{\frac{\delta |X_1|}{2}}$$

where $C = 3! \cdot \frac{2^3}{\delta^3}$ (where does this come from)? Hence

$$\begin{aligned}
 \mathbb{E} \left[|\theta X_1|^3 e^{\frac{\delta |X_1|}{2}} \right] &\leq C |\theta|^3 \mathbb{E} \left[e^{\frac{\delta |X_1|}{2}} \cdot e^{\frac{\delta |X_1|}{2}} \right] \\
 &= C |\theta|^3 \mathbb{E} \left[e^{\delta |X_1|} \right] \\
 &\leq C |\theta|^3 \mathbb{E} \left[e^{\delta X_1} + e^{-\delta X_1} \right]
 \end{aligned}$$

so that

$$\left| \mathbb{E} \left[\sum_{k \geq 3} \frac{\theta^k}{(\sqrt{n})^k} \cdot \frac{1}{k!} X_1^k \right] \right| \leq C' \cdot |\theta|^3$$

Hence the LHS is $O(|\theta|^3)$, so $o(|\theta|^2)$ as required. Now

$$\begin{aligned}
 m \left(\frac{\theta}{\sqrt{n}} \right) &= 1 + \frac{\theta^2}{2n} + o \left(\frac{\theta^2}{n} \right) \\
 \implies \left(m \left(\frac{\theta}{\sqrt{n}} \right) \right)^n &\rightarrow e^{\frac{\theta^2}{2}}
 \end{aligned}$$

as $n \rightarrow \infty$ (aren't we implicitly taking $\theta \rightarrow 0$ here? how can this be valid for any $\theta \neq 0$ then? since we take $\theta \rightarrow 0$ for little o of $|\theta|^2$ as well as $n \rightarrow \infty$). $\blacksquare$

Example 10.10

Suppose $S_n \sim \text{Bin}(n, p)$. We can write

$$S_n = X_1 + \cdots + X_n$$

where the X_i are i.i.d. Bernoulli random variables with parameter p . Hence, inserting the mean and variance of the Bernoulli distribution, we obtain

$$\frac{S_n - np}{\sqrt{np(1-p)}} \xrightarrow{(d)} \mathcal{N}(0, 1)$$

so that

$$S_n \approx \mathcal{N}(np, np(1-p))$$

for large n .

Example

Suppose $S_n \sim \text{Poi}(n)$, so that

$$S_n = X_1 + \cdots + X_n$$

where the X_i are i.i.d. distributed $\text{Poi}(1)$. Subbing in the mean and variance of the Poisson distribution, we obtain

$$\begin{aligned} \frac{S_n - n}{\sqrt{n}} &\xrightarrow{(d)} \mathcal{N}(0, 1) \\ \implies S_n &\approx \mathcal{N}(n, n) \end{aligned}$$

Lecture 24 – 18/03/26

Suppose that, in a referendum, every individual votes yes with probability p and no with probability $1 - p$, independently of one another, and that we want to estimate p by sampling.

Let X_i be 1 if the i^{th} individual sampled voted ‘yes’, 0 otherwise. Let $S_N = X_1 + \cdots + X_n$, and let $\hat{p}_N = \frac{S_N}{N}$, which goes to p a.s. as $n \rightarrow \infty$ by the strong law of large numbers – in particular, $S_N \sim \text{Bin}(n, p)$. For a given desired error, how large should N be?

By the CLT,

$$\begin{aligned} S_N &\approx Np + \sqrt{Np(1-p)}Z \\ \implies \hat{p}_N &\approx p + \frac{\sqrt{p(1-p)}}{\sqrt{N}}Z \\ \implies |\hat{p}_N - p| &\approx \frac{\sqrt{p(1-p)}}{\sqrt{N}}|Z| \end{aligned}$$

Hence, we want to solve

$$\mathbb{P}\left(\frac{\sqrt{p(1-p)}}{\sqrt{N}}|Z| \geq \varepsilon\right) \leq p'$$

We can solve this using Φ , and we can avoid worrying about not knowing what p is by simply using the fact that $\sqrt{p(1-p)} \leq \frac{1}{\sqrt{2}}$.

§11 Simulation of random variables

Suppose one can generate uniform random numbers in $[0, 1]$. How can we sample some other random variable?

Suppose first F is one-to-one – then set $X = F^{-1}(U)$, so

$$\mathbb{P}(X \leq x) = \mathbb{P}(F^{-1}(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x)$$

If F is not one-to-one, we define its *generalised inverse*

$$G(u) = \inf\{x \in \mathbb{R} : u \leq F(x)\}$$

Lemma 11.1

$G(u) \leq x$ if and only if $u \leq F(x)$.

Proof. The reverse implication is immediate. For the forward implication, observe there exists $x_n \geq G(u)$ such that $x_n \rightarrow G(u)$ from above and $u \leq F(x_n)$. By right continuity of F , $F(x_n) \rightarrow F(G(u))$ as $n \rightarrow \infty$, so $u \leq F(G(u))$. Hence, if $G(u) \leq x$, then $u \leq F(G(u)) \leq F(x)$ as F is increasing. ■

Now, we can set $X = G(U)$, so that

$$\mathbb{P}(X \leq x) = \mathbb{P}(G(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x)$$

§11.1 Box-Muller transform

We would like to simulate X and Y independent $\mathcal{N}(0, 1)$ using two independent $U[0, 1]$ random variables, U and V . Theoretically, we can just apply Φ^{-1} to each of the uniform random variables, but indeed this does not have a closed form.

Consider making a complex number $X + iY$ from X and Y – we have already shown that, for this complex number, R has density $re^{-\frac{r^2}{2}}$, $\theta \sim U[0, 2\pi]$, and these are independent. Now put

$$\begin{aligned}\theta &= 2\pi U \\ R &= \sqrt{-2 \log V}\end{aligned}$$

where the R expression comes from the inverse of the CDF for R . Indeed, we have

$$\mathbb{P}(R \geq r) = \mathbb{P}\left(\sqrt{-2 \log V} \geq r\right) = \mathbb{P}\left(V \leq e^{-\frac{r^2}{2}}\right) = e^{-\frac{r^2}{2}}$$

so R has the correct cdf.

§11.2 Rejection sampling

Suppose we have an r.v. X with density

$$f(x) = \frac{1(x \in A)}{|A|}$$

where $A \in [0, 1]^d$ has volume $|A|$. We can sample uniformly in A by sampling coordinates uniformly, and discarding samples until we end up in A .

Let $N = \min\{n : U_n \in A\}$, and $X = U_N$ – we need to show X has density f . It suffices to show that

$$\mathbb{P}(X \in B) = \frac{|B \cap A|}{|A|}$$

We have

$$\begin{aligned} \mathbb{P}(X \in B) &= \sum_{n=1}^{\infty} \mathbb{P}(U_n \in B, N = n) \\ &= \sum_{n=1}^{\infty} \mathbb{P}(U_n \in B, U_1 \notin A, \dots, U_{n-1} \notin A, U_n \in A) \\ &= \sum_{n=1}^{\infty} \mathbb{P}(U_n \in B \cap A) \mathbb{P}(U_1 \notin A)^{n-1} \\ &= \sum_{n=1}^{\infty} |B \cap A| (1 - |A|)^{n-1} \\ &= \frac{|B \cap A|}{|A|} \end{aligned}$$